

## NOTES

IDLE

Hello!!!

This note is mainly used to record some simple but interesting problems that I encounter in daily life.

*For as the heavens are higher than the earth,  
so are my ways higher than your ways,  
and my thoughts than your thoughts.*

— Isaiah 55:9



FIGURE 1. Night City

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<sup>1</sup>This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre.

1. 2023.12.22

**1.1. Identification of co-character groups with Lie algebras.** Settings:  $G$  complex connected reductive algebraic group.  $Q \subseteq P \subseteq \mathfrak{h}^*$  are root lattice and weight group, respectively.  $\check{Q} \subseteq \check{P} \subseteq \mathfrak{h}$  are coroot lattice and coweight group, respectively.  $H$  is a maximal torus of  $G$ ,  $\mathfrak{h}$  is its Lie algebra.

There are identifications

$$\begin{aligned} I: \quad X_*(H) \otimes \mathbb{C} &\longrightarrow \mathfrak{h} \\ \phi &\longmapsto d\phi(1) \\ \\ J: \quad X^*(H) \otimes \mathbb{C} &\longrightarrow \mathfrak{h}^* \\ \varphi &\longmapsto d\varphi \end{aligned}$$

Under these identifications we have

$$\begin{aligned} \check{P} &= \{\check{\lambda} \in \mathfrak{h} \mid \langle \alpha, \check{\lambda} \rangle \in \mathbb{Z}, \text{ for all } \alpha \in Q\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}\langle \alpha, \check{\lambda} \rangle) = 1, \text{ for all } \alpha \in Q\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \alpha(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{ for all } \alpha \in Q\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}\check{\lambda}) \in Z(G)\}. \end{aligned}$$

And

$$\begin{aligned} X_*(H) &= \{\check{\lambda} \in \mathfrak{h} \mid d\phi(\check{\lambda}) \in \mathbb{Z}, \text{ for all } \phi \in X^*(H)\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}d\phi(\check{\lambda})) = 1, \text{ for all } \phi \in X^*(H)\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \phi(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{ for all } \phi \in X^*(H)\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}\check{\lambda}) = 1\}. \end{aligned}$$

Similarly

$$P = \{\lambda \in \mathfrak{h}^* \mid \exp(2\pi\sqrt{-1}\lambda) \in Z(\check{G})\}.$$

$$X^*(H) = \{\lambda \in \mathfrak{h}^* \mid \exp(2\pi\sqrt{-1}\lambda) = 1\}.$$

## 1.2. Pinnings of algebraic groups.

**Theorem 1.1** (see AdC18). *Inner automorphism group  $\text{Inn}(G)$  of  $G$  is equal to  $G/Z(G)$ ,  $\text{Inn}(G)$  acts freely and transitively on the set of pinnings.*

I know that there is an analogy in compact connected groups, and it has been proved.

**Theorem 1.2.** *If  $G$  is a compact connected group, then its inner automorphism group  $\text{Inn}(G) = G/Z(G)$  acts freely and transitively on the set of pinnings. (where pinnings for it is a bit different from the complex case, that  $(T, B_{\mathbb{C}}, \mathcal{X})$  is a pinning if  $T$  is a maximal torus,  $B_{\mathbb{C}}$  a Borel of  $G_{\mathbb{C}}$  containing  $T$ ,  $\mathcal{X}$  is a set of real rays in simple root space.)*

*Proof.* Here's proof from mathoverflow [Lspise's answer](#):

To prove this, I'll use a few pieces of structure theory:

1. All maximal tori in  $G$  are  $G^\circ$ -conjugate.
2. All Borel subgroups of  $G_{\mathbb{C}}$  are  $G_{\mathbb{C}}^\circ$ -conjugate.
3. For every maximal torus  $T$  in  $G$ , the map  $W(G^\circ, T) \rightarrow W(G_{\mathbb{C}}^\circ, T_{\mathbb{C}})$  is an isomorphism.
4. If  $G_{\text{sc}}$  and  $(G_{\mathbb{C}})_{\text{sc}}$  are the simply connected covers of the derived groups of  $G^\circ$  and  $G_{\mathbb{C}}^\circ$ , then  $(G_{\text{sc}})_{\mathbb{C}}$  equals  $(G_{\mathbb{C}})_{\text{sc}}$ .
5. [Every compact Lie group has a finite subgroup that meets every component.](#)

I only need (4) to prove that, for every maximal torus  $T$  in  $G$ , the map from  $T$  to the set of conjugation-fixed elements of  $T/Z(G^\circ)$  is surjective. This is probably a well known fact in its own right for real-group theorists.

Now consider triples  $(T, B_{\mathbb{C}}, \mathcal{X})$  as follows:  $T$  is a maximal torus in  $G$ ;  $B_{\mathbb{C}}$  is a Borel subgroup of  $G_{\mathbb{C}}^\circ$  containing  $T_{\mathbb{C}}$ , with a resulting set of simple roots  $\Delta(B_{\mathbb{C}}, T_{\mathbb{C}})$ ; and  $\mathcal{X}$  is a set consisting of a real ray in each complex simple root space (i.e., the set of positive real multiples of some fixed non-0 vector). (Sorry about the pair of modifiers "complex simple".) I will call these 'pinnings', although it doesn't agree with the usual terminology (where we pick individual root vectors, not rays). I claim that  $G^\circ/Z(G^\circ)$  acts simply transitively on the set of pinnings.

Once we have transitivity, freeness is clear: if  $g \in G^\circ$  stabilises some pair  $(T, B_C)$ , then it lies in  $T$ , and so stabilises every complex root space; but then, for it to stabilise some choice of rays  $\mathcal{X}$ , it has to have the property that  $\alpha(g)$  is positive and real for each simple root  $\alpha$ ; but also  $\alpha(g)$  is a norm-1 complex number, hence trivial, for each simple root  $\alpha$ , hence for each root  $\alpha$ , so that  $g$  is central.

For transitivity, since (1) all maximal tori in  $G$  are  $G^\circ$ -conjugate, so (2) for every maximal torus  $T$  in  $G$ , the Weyl group  $W(G_C^\circ, T_C)$  acts transitively on the Borel subgroups of  $G_C^\circ$  containing  $T_C$ , and (3)  $W(G^\circ, T) \rightarrow W(G_C^\circ, T_C)$  is an isomorphism, it suffices to show that all possible sets  $\mathcal{X}$  are conjugate. Here's the argument that I came up with to show that they are even  $T$ -conjugate; I think it can probably be made much less awkward. Fix a simple root  $\alpha$ , and two non-0 elements  $X_\alpha$  and  $X'_\alpha$  of the corresponding root space. Then there are a positive real number  $r$  and a norm-1 complex number  $z$  such that  $X'_\alpha = rzX_\alpha$ . Choose a norm-1 complex number  $w$  such that  $w^2 = z$ . There is then a unique element  $s_{\text{ad}}$  of  $T_C / Z(G_C^\circ)$  such that  $\alpha(s_{\text{ad}}) = w$ , and  $\beta(s_{\text{ad}}) = 1$  for all simple roots  $\beta \neq \alpha$ . By (4), we can choose a lift  $s_{\text{sc}}$  of  $s_{\text{ad}}$  to  $(G_{\text{sc}})_C = (G_C)_{\text{sc}}$ , which necessarily lies in the preimage  $(T_C)_{\text{sc}}$  of (the intersection with the derived subgroup of)  $T$ , and put  $t_{\text{sc}} = s_{\text{sc}} \cdot \overline{s_{\text{sc}}}$ . Then

$$\alpha(t_{\text{sc}}) = \alpha(s_{\text{sc}})\overline{\alpha(s_{\text{sc}})} = \alpha(s_{\text{sc}})\overline{\alpha(s_{\text{sc}})^{-1}} = w \cdot \overline{w^{-1}} = z,$$

and, similarly,  $\beta(t_{\text{sc}}) = 1$  for all simple roots  $\beta \neq \alpha$ . Now the image  $t$  of  $t_{\text{sc}}$  in  $G_C^\circ$  lies in  $T_C$  and is fixed by conjugation, hence lies in  $T$ ; and  $\text{Ad}(t)X_\alpha = zX_\alpha$  lies on the ray through  $X'_\alpha$ .

Since  $G$  also acts on the set of pinnings, we have a well defined map  $p : G \rightarrow G^\circ / Z(G^\circ)$  that restricts to the natural projection on  $G^\circ$ . Now  $\ker(p)$  meets every component, but it contains  $Z(G^\circ)$ , so it need not be finite. Applying (5) to the Lie group  $\ker(p)$  yields the desired subgroup  $H$ . Note that, as requested in your [improved classification](#), conjugation by any element of  $H$  fixes a pinning, hence, if inner, must be trivial.

□

**Remark 1.3.** The pinnings show that split reductive algebraic groups look like butterflies!

According to Milne, Grothendieck used the following analogy to talk about the structure theory of algebraic groups. An algebraic group is like a butterfly. The body of the butterfly is the maximal torus. The wings are two opposite Borel subgroups. And the pins are the pinning maps.

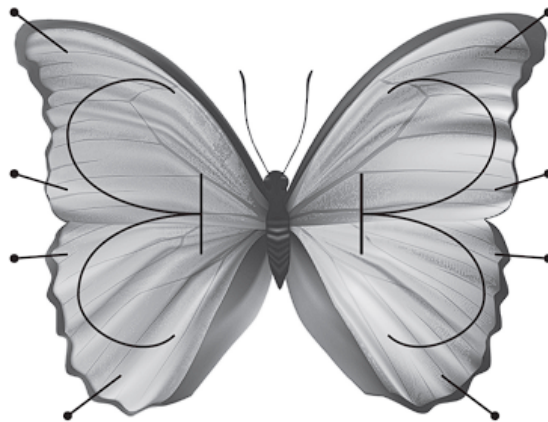


FIGURE 2. Pinning of butterfly



**1.3. Coherent continuation representations.** Until the end of today,  $G$  denotes a real reductive group in Harish-Chandra class.

Let's recall some basic definitions of coherent continuation representations, in order for me to do calculations in the case of real classical groups.

Fix a connected reductive complex Lie group  $G_{\mathbb{C}}$ , together with a Lie group homomorphism  $\iota : G \rightarrow G_{\mathbb{C}}$  such that its differential  $d\iota : \text{Lie}(G) \rightarrow \text{Lie}(G_{\mathbb{C}})$  has the following two properties:

- (1) the kernel of  $d\iota$  is contained in the center of  $\text{Lie}(G)$ ;
- (2) the image of  $d\iota$  is a real form of  $\text{Lie}(G)$ .

The analytical weight lattice of  $G_{\mathbb{C}}$  is identified with a subgroup of  ${}^a\mathfrak{h}^*$  via  $d\iota$ . We write  $Q_{\iota} \subset {}^a\mathfrak{h}^*$  for this subgroup. Denote root lattice by  $Q_{\mathfrak{g}}$ , weight lattice by  $Q^{\mathfrak{g}}$ , then

$$Q_{\mathfrak{g}} \subset Q_{\iota} \subset Q^{\mathfrak{g}}.$$

They are all  $W$  stable subgroups. And we have some categories:

- (1)  $\text{Rep}(\mathfrak{g}, Q_{\iota})$  the category of all finite-dimensional representations of  $\mathfrak{g}$  whose weight are contained in  $Q_{\mathfrak{g}}$ , **it can be identified with  $\text{Rep}(\text{Lie}(G_{\mathbb{C}}), Q_{\iota})$  and hence  $\text{Rep}_{\text{finite}}(G_{\mathbb{C}})$** ;
- (2)  $\text{Rep}(G)$  the category of all Casselman-Wallach representations of  $G$ , and its full subcategories  $\text{Rep}_{\mu}(G)$ ,  $\text{Rep}_{\mathbb{S}}(G)$ .

To denote their Grothendieck groups, we replace  $\text{Rep}$  by  $\mathcal{K}$ .

Fix a  $Q_{\iota}$ -coset  $\Lambda = \lambda + Q_{\iota} \subset {}^a\mathfrak{h}^*$ .

**Definition 1.4** (coherent family). A  $\mathcal{K}(G)$ -valued  $\Lambda$  coherent family is a map

$$\Phi : \Lambda \rightarrow \mathcal{K}(G)$$

satisfies the following two conditions:

- (1) for all  $\nu \in \Lambda$ ,  $\Phi(\nu) \in \mathcal{K}_{\nu}(G)$ ;
- (2) for all representations  $F \in \text{Rep}(\mathfrak{g}, Q_{\iota})$  and all  $\nu \in \Lambda$ ,

$$F \cdot (\Phi(\nu)) = \sum_{\mu} \Phi(\nu + \mu),$$

where  $\mu$  runs over all weights of  $F$ , counted with multiplicities.

The set of coherent families is denoted by  $\text{Coh}_{\Lambda}(\mathcal{K}(G))$

We have the integral Weyl group  $W_{\Lambda} = \{w \in W \mid w\lambda - \lambda \in Q_{\iota}\}$  there is a representation of  $W_{\Lambda}$  on  $\text{Coh}_{\Lambda}(\mathcal{K}(G))$  by

$$(w \cdot \Phi)(\nu) = \Phi(w^{-1}\nu).$$

This is called a coherent continuation representation.

2. 2023.12.23

**2.1. Review of regular parameter.** Following yesterday's discussion of coherent continuation representations, we define two basis for it, use Langland classification.

Suppose  $H$  is a Cartan subgroup of  $G$ , by which we mean a subgroup that is the centralizer of a Cartan subalgebra of  $\text{Lie}(G)$  (In the case of linear groups, it's just a real algebraic torus.).  $H$  has a unique maximal compact subgroup (since  $G$  is in Harish-Chandra's class, adjoint representation of  $H$  is trivial,  $H_0$  lies in the center of  $H$ , as we know, all maximal subgroups of Lie groups are conjugate to each other, use Cartan decomposition of  $H$ , we win. In fact,  $H$  has a unique maximal compact torus and a unique maximal split torus) We denote by  $\Delta_{\mathfrak{h}} \subset \mathfrak{h}^*$  the root system of  $\mathfrak{g}$ . A root is called imaginary if  $\check{\alpha} \in \mathfrak{t}$  (i.e. roots which take pure imaginary values on  $\text{Lie}(H)$ ), an imaginary root  $\alpha$  is called compact if  $\mathfrak{g}_{\alpha}$  and  $\mathfrak{g}_{-\alpha}$  is contained in the complexified Lie algebra of a common compact subgroup of  $G$ . (this is equivalent to  $\mathfrak{g}_{\alpha}$  belong to a Lie subalgebra of a compact subgroup, since  $\mathfrak{g}_{\alpha}$  lies in a Lie subalgebra of maximal compact subgroup is equivalent to  $\mathfrak{g}_{-\alpha}$  Lies in the same subgroup)

There are two important facts about the representation theory of  $H$ :

- (1) Every Casselman-Wallach representation of  $H$  is finite-dimensional, it follows directly from that  $H$  is an extension of an abelian group and a finite group;
- (2) For every  $\Gamma \in \text{Irr}(H)$  differential of  $\Gamma$  is a direct sum of one-dimensional representations attached to a unique  $d\Gamma \in \mathfrak{h}^*$ . (it is true because  $H$  is in Harish-Chandra's class).

For every Borel subalgebra  $\mathfrak{b}$  of  $\mathfrak{g}$  containing  $\mathfrak{h}$ , write

$$\xi_{\mathfrak{b}} : \mathfrak{h} \rightarrow {}^a\mathfrak{h},$$

for the linear isomorphism attached to  $\mathfrak{b}$  defined by

$$\mathfrak{h} \hookrightarrow \mathfrak{b} \rightarrow \mathfrak{b}/[\mathfrak{b}, \mathfrak{b}] = {}^a\mathfrak{h}.$$

The transpose inverse of this map is still denoted by  $\xi_{\mathfrak{b}} : {}^a\mathfrak{h}^* \rightarrow \mathfrak{h}^*$ .

Write

$$W({}^a\mathfrak{h}^*, \mathfrak{h}^*) := \{\xi_{\mathfrak{b}} : {}^a\mathfrak{h}^* \rightarrow \mathfrak{h}^* \mid \mathfrak{b} \text{ is a Borel subalgebra containing } \mathfrak{h}\}.$$

**Definition 2.1.** For every element  $\xi \in W({}^a\mathfrak{h}^*, \mathfrak{h}^*)$ , put

$$\delta(\xi) := \frac{1}{2} \cdot \sum_{\substack{\alpha \text{ is an imaginary root in } \xi\Delta^+}} \alpha - \sum_{\substack{\beta \text{ is an compact imaginary root in } \xi\Delta^+}} \beta \in \mathfrak{h}^*.$$

Write  $\mathcal{P}_{\Lambda}(G)$  for the set of all triples  $\gamma = (H, \xi, \Gamma)$ , where  $H$  is a Cartan subgroup of  $G$ ,  $\xi \in W({}^a\mathfrak{h}^*, \mathfrak{h}^*)$ , and

$$\begin{aligned} \Gamma : \Lambda &\longrightarrow \text{Irr}(H) \\ \nu &\longmapsto \Gamma_{\nu} \end{aligned}$$

is a map with the following properties:

- (1)  $\Gamma_{\nu+\beta} = \Gamma_{\nu} \otimes \xi(\beta)$  for all  $\beta \in Q_i$  and  $\nu \in \Lambda$ ;
- (2)  $d\Gamma_{\nu} = \xi(\nu) + \delta(\xi)$  for all  $\nu \in \Lambda$ . (note that Cartan subgroups not always have irr repn with this differential!)

Here  $\xi(\beta)$  is naturally viewed as a character of  $H$  by using the homomorphism  $\iota : H \rightarrow H_{\mathbb{C}}$ , and  $H_{\mathbb{C}}$  is a Cartan subgroup of  $G_{\mathbb{C}}$  containing  $\iota(H)$ .



## 2.2. Outer automorphism of algebraic groups.

**Lemma 2.2** (a simple discover of abstract algebra).  *$G$  an abstract group,  $H$  a normal subgroup of  $G$ , if  $G$  acts transitively on a set  $X$  such that the restriction of this action to  $H$  is free and transitive, then the short exact sequence of groups*

$$1 \rightarrow H \rightarrow G \rightarrow G/H \rightarrow 1$$

*Proof.* Actually choose an element  $x \in X$ , we have

$$G \simeq \text{Stab}_G(x) \ltimes H.$$

□

Using this lemma, we can see that for a complex connected reductive algebraic group  $G$ ,  $\text{Aut}(G)$  acts on the set of pinning transitivity, with its normal subgroup the  $\text{Inn}(G)$  acts freely and transitively, so every pinning gives us a split of the short exact sequence

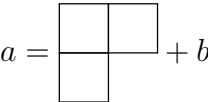
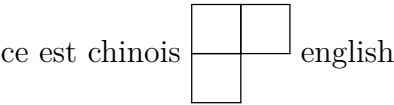
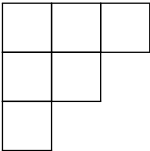
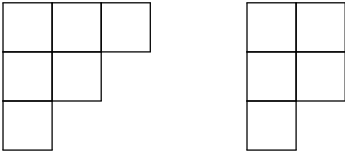
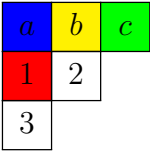
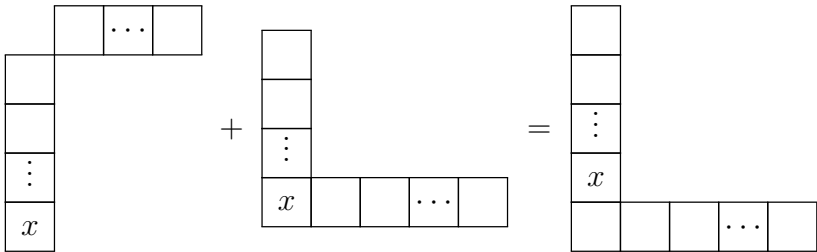
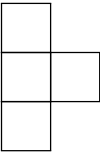
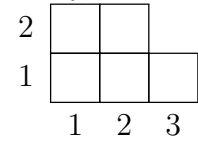
$$1 \rightarrow \text{Inn}(G) \rightarrow \text{Aut}(G) \rightarrow \text{Out}(G) \rightarrow 1.$$

Some facts about split connected reductive algebraic groups over arbitrary field  $k$ :

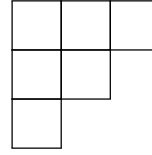
- (1)  $\{\text{split connected reductive algebraic groups}\}/\sim = \{\text{root datums}\}/\sim$ ;
- (2) For any two automorphisms of  $G$ , if they induce the same automorphism of root datum, then they differ by an inner automorphism;
- (3) Fix a pinning, we get an isomorphism  $\text{Out}(G) \simeq \text{Aut}(D_b)$ .

3. 2024.2.27

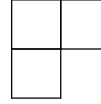
Today, I learned how to draw the Young tableau.



youngdiagram1:



young2:



For a compact connected Lie group  $G$ , choose a maximal torus  $T$ , there is a canonical 1-1 correspondence:

$$X^*(T)/W \longleftrightarrow (X^*(T) + \rho)/W \longleftrightarrow \text{Irr}(G)$$

Moreover, if we choose a positive system  $\Psi^+$ , and then  $\rho_{\Psi^+}$ , the above sets also canonically isomorphic to the set of dominant analytically integral weights:

$$\{\lambda \in X^*(T) \mid \langle \lambda, \alpha \rangle \geq 0, \text{ for all } \alpha \in \Psi^+\}$$

This can be generalized to parameter of discrete series representations for connected semisimple Lie groups  $G$  with compact Cartan subgroup  $T$  :

$$(X^*(T) + \rho)^{reg}/W_c$$

If a discrete series representation  $V$  has minimal  $K$  type which has maximal weight  $\nu$ , then it's parameter is  $\lambda = \nu - \rho + 2\rho_c$ . (not sure)

Note that for a fixed infinitesimal character, there are exactly  $|W/W_c|$  different discrete series representations.

## 4. 2024.2.28

Some facts about continuous complex characters of some common abelian topological groups:

| field                           | character group                | correspondence                                           |
|---------------------------------|--------------------------------|----------------------------------------------------------|
| $(\mathbb{R}_+^\times, \times)$ | $\mathbb{C}$                   | $\xi \mapsto (t \mapsto t^\xi)$                          |
| $(\mathbb{C}^\times, \times)$   | $\mathbb{C} \times \mathbb{Z}$ | $(\xi, n) \mapsto (z \mapsto  z ^\xi (\frac{z}{ z })^n)$ |
| $(\mathbb{S}^1, \times)$        | $\mathbb{Z}$                   | $n \mapsto (z \mapsto z^n)$                              |
| $(\mathbb{Q}_p, +)$             | $\mathbb{Q}_p$                 | $\xi \mapsto (x \mapsto \exp(2\pi i \xi \{x\}_p))$       |

**Remark 4.1.** Note that all continuous characters of  $\mathbb{Q}_p$  are automatically locally constant (smooth), which is consistent with the case of real lie groups.

5. 2024.10.22

**Lemma 5.1.** *The associated variety of an irreducible Casselman-Wallach representation of a real reductive group in Harish-Chandra class is the closure of a nilpotent orbit in  $\mathfrak{g}^*$ .*

*Proof.* By the work of Borho, Brylinski [BB82], and Joseph [Jos85], the associated variety of any primitive ideal (annihilator of an irreducible  $\mathfrak{g}$ -module) is the closure of a single nilpotent orbit. So we only need to prove that  $\text{gr}(\text{Ann}_{\mathcal{U}(\mathfrak{g})}(V))$  is a primitive ideal.

Since the  $K$ -finite part  $V_K$  is dense in  $V$ ,  $\text{Ann}_{\mathcal{U}(\mathfrak{g})}(V) = \text{Ann}_{\mathcal{U}(\mathfrak{g})}(V_K)$ , we only need to consider the annihilator of the irreducible  $(\mathfrak{g}, K)$ -module  $V_K$ .

If  $G$  is connected, then  $K$  is also connected, so  $V_K$  is irreducible as  $\mathcal{U}(\mathfrak{g})$ -module, in this case,  $\text{Ann}_{\mathcal{U}(\mathfrak{g})}(V)$  is a primitive ideal.

If  $G$  is not connected, let  $G_0$  be the identity component of  $G$ ,  $K_0 = G_0 \cap K$  is a maximal compact subgroup of  $G_0$ , then  $V_K|_{G_0}$  is a finite length  $(\mathfrak{g}, K_0)$ -module, it has an irreducible quotient  $V_K \rightarrow V_0$ , then according to the theory of induction of  $(\mathfrak{g}, K)$ -modules [KV95, Chapter 2] we have an injective morphism of  $(\mathfrak{g}, K_0)$ -modules:

$$(5.1) \quad V_K|_{K_0} \hookrightarrow (\text{induced}_{K_0}^K(V_0))|_{K_0} \cong \bigoplus_{\text{double cosets } K_0 k K_0} kV_0,$$

where  $kV_0$  be the  $(\mathfrak{g}, K_0)$ -module whose underlying space is  $V_0$ , whose action by  $X \in \mathfrak{g}$  is the usual action by  $\text{Ad}(k)X$ , whose action by  $g \in K_0$  is the usual action by  $k^{-1}gk$ , it has the same annihilator in  $\mathcal{U}(\mathfrak{g})$  as  $V_0$  since  $G$  is in Harish-Chandra's class, and hence the annihilator of  $V_K$  is the same as the annihilator of  $V_0$ , which is a primitive ideal.

□

6. 2024.11.28

Any split connected reductive group over arbitrary field can be uniquely defined and split over  $\mathbb{Z}$ . (Chevalley group)

Classification of split reductive group is the same over arbitrary non-empty schemes, which is equivalent to the classification of root data.(cf. [SGA3](#))



7. 2024.12.17

**Geometric Pre-quantization**

Let  $X$  be a smooth manifold. There are natural maps

$$\text{Pic}(X) \longrightarrow H_{dR}^2(X; \mathbb{R}) \simeq H_{Cech}^2(X; \mathbb{R}).$$

Construction of the first map: For a (complex) line bundle  $\mathbb{L} \in \text{Pic}^h(X)$ , let  $\langle, \rangle$  be a Hermitian metric on it and  $\nabla$  be a unitary connection. (Every line bundle  $\mathbb{L} \in \text{Pic}(X)$  has a Hermitian metric and a unitary connection on it.)

- choose a locally trivialization  $\{U_\alpha\}$  of  $\mathbb{L}$ , such that all  $U_\alpha$  and  $U_\alpha \cap U_\beta$  are contractible;
- choose Unitary local frames  $\{s_\alpha\}$  for  $\mathbb{L}$  on each  $U_\alpha$ , then  $s \leftrightarrow (f_\alpha)$  and  $\nabla \leftrightarrow (\theta_\alpha)$  (unitary implies that  $\theta_\alpha$  are pure imaginary);
- define  $\Omega := d\theta_\alpha$ , and  $c_1(\mathbb{L}) := [\frac{1}{2\pi i}\Omega] \in H_{dR}^2(X; \mathbb{R})$  ( $\Omega = \text{curv}\nabla$ ).

$c_1(\mathbb{L})$  is independent of the choice of  $\langle, \rangle$  and  $\nabla$  by Chern-Weil theory. And we can see that  $\theta_\alpha - \theta_\beta = \frac{dg_{\alpha,\beta}}{g_{\alpha,\beta}} = d\log(g_{\alpha,\beta})$  on  $U_\alpha \cap U_\beta$ , where  $g_{\alpha,\beta}$  is the transition function ( $s_\beta = g_{\alpha,\beta}s_\alpha$ ).

Construction of the second map (classical de-Rham isomorphism), choose a 2-form  $\omega \in H_{dR}^2(X; \mathbb{R})$

- choose a cover  $\{U_\alpha\}$  of  $\mathbb{L}$ , such that all  $U_\alpha$  and  $U_\alpha \cap U_\beta$  are contractible;
- on each  $U_\alpha$ , choose a 1-form  $\theta_\alpha$  such that  $d\theta_\alpha = \omega|_{U_\alpha}$ ;
- since  $d\theta_\alpha = d\theta_\beta$  on  $U_\alpha \cap U_\beta$ , we can choose smooth function  $c_{\alpha,\beta}$  such that  $dc_{\alpha,\beta} = \theta_\alpha - \theta_\beta$ ;
- let  $c_{\alpha,\beta,\gamma} = c_{\alpha,\beta} + c_{\beta,\gamma} + c_{\gamma,\alpha}$ , which is a constant, and we get the corresponding 2-cocycle  $(c_{\alpha,\beta,\gamma}) \in H_{Cech}^2(X; \mathbb{R})$ .

If  $\omega$  come from the first map, we can choose  $c_{\alpha,\beta} = \frac{1}{2\pi i} \log g_{\alpha,\beta}$ , and hence  $c_{\alpha,\beta,\gamma} = \frac{1}{2\pi i} (\log g_{\alpha,\beta} + \log g_{\beta,\gamma} + \log g_{\gamma,\alpha}) \in \mathbb{Z}$ . So the morphism is in fact

$$\text{Pic}(X) \longrightarrow H_{dR}^2(X; \mathbb{Z}) \simeq H_{Cech}^2(X; \mathbb{Z}).$$

In fact the first map is also an isomorphism. (cf. [prequantization](#)) For a 2-form  $\omega$ , choose  $\theta_\alpha$  and  $c_{\alpha,\beta}$  as above, let  $g_{\alpha,\beta} = \exp(2\pi i c_{\alpha,\beta})$ , these are transition functions of a complex line bundle  $\mathbb{L}$ .

**Corollary 7.1.** *Assume that  $G$  is a connected Lie group. The following are equivalent:*

- $\Omega \subseteq \mathfrak{g}^*$  is integral ( $[\sigma_\Omega] \in H^2(X; \mathbb{Z})$ );
- there exists a  $G$ -equivariant complex line bundle over  $\Omega$  with a  $G$ -invariant Hermitian connection  $\nabla$  such that

$$\frac{1}{2\pi i} \text{curv}(\nabla) = \sigma_\Omega.$$

**Theorem 2.6** (Weil). *Let  $M$  be a smooth manifold and  $\omega$  a real, closed 2-form whose cohomology class  $[c]$  is integral. Then there is a unique Hermitian line bundle  $\mathbb{L}$  over  $M$  with unitary connection  $\nabla$  so that  $c_1(\mathbb{L}) = [c]$ .*

*Sketch of Proof.* Existence: Reverse the arguments above. Since  $[c]$  is integral,  $e^{2\pi i c_{ijk}} = 1$ . As a consequence,  $g_{ij}g_{jk}g_{kl} = 1$ . So  $g_{ij}$ 's are the transition function for some line bundle whose first Chern class is  $[c]$ .

Uniqueness: Suppose  $c_1(\mathbb{L}) = c_1(\tilde{\mathbb{L}})$ . Let  $h_{ij} = \frac{1}{2\pi i} g_{ij}$  and define  $\tilde{h}_{ij}$  similarly. Then the functions  $\hat{h}_{ij} = h_{ij} - \tilde{h}_{ij}$  satisfies the relation

$$\hat{h}_{ij} + \hat{h}_{jk} + \hat{h}_{ki} = 0.$$

We take a partition of unity  $\rho_k$  and let  $\lambda_i = e^{2\pi i \sum \hat{h}_{ki} \rho_k}$ . Then

$$\lambda_i g_{ij} \lambda_j^{-1} = e^{2\pi i (h_{ij} + \sum (\hat{h}_{ki} - \hat{h}_{kj}) \rho_k)} = e^{2\pi i (h_{ij} - \hat{h}_{ij})} = \tilde{g}_{ij}.$$

So as line bundles  $\mathbb{L} \simeq \tilde{\mathbb{L}}$ . □

FIGURE 3. Proof

8. 2025.8.14<sup>2</sup>

Let  $G$  be a finite central extension of an open subgroup of  $\mathbb{R}$ -points of a connected reductive algebraic group (reductive Nash group [Sun15]) with Lie algebra  $\mathfrak{g}$ . Fix an invariant nondegenerate bilinear form  $B(-, -)$  on  $\mathfrak{g}$ .

For an irreducible Casselman-Wallach representation  $\pi$  of  $G$ , let  $\text{Wh}_{\mathcal{O}}(\pi)$  be the space of generalized Whittaker models of  $\pi$  associated to the nilpotent orbit  $\mathcal{O}$ . Let

$$\text{WO}(\pi) := \{\mathcal{O} \mid \mathcal{O} \text{ is a nilpotent orbit such that } \text{Wh}_{\mathcal{O}}(\pi) \neq 0\},$$

and the *Whittaker support*  $\text{WS}(\pi)$  of  $\pi$  be the set of all maximal orbits in  $\text{WO}(\pi)$  with respect to closure inclusion:

$$\text{WS}(\pi) := \{\mathcal{O} \mid \mathcal{O} \text{ is a maximal orbit in } \text{WO}(\pi)\}.$$

Another nilpotent invariant is the wavefront cycle:

$$\mathcal{WF}(\pi) = \sum_i m_i [\mathcal{O}_i].$$

With the corresponding wavefront set :

$$\text{WF}(\pi) = \bigcup_{m_i \neq 0} \overline{\mathcal{O}_i}.$$

Mœglin-Waldspurger [MW87] proved that  $\text{WF}(\pi) = \text{WS}(\pi)$  when  $F$  is a  $p$ -adic field. We want to test this conjecture for arbitrary reductive almost linear Nash groups.

**Conjecture 8.1** (Main conjecture). *For any reductive almost linear Nash group  $G$  and any irreducible Casselman-Wallach representation  $\pi$ :*

$$\text{WF}(\pi) = \text{WS}(\pi).$$

Take a nilpotent orbit  $\mathcal{O} \subseteq \mathfrak{g}$  (or  $\mathfrak{sl}_2$ -triple  $\gamma = \{Y, H, X\}$  in  $\mathfrak{g}$ ).  $\mathfrak{g}/\mathfrak{g}^Y$  is a symplectic space with symplectic form  $\langle -, - \rangle := B(Y, [-, -])$ . Set  $G^Y := \text{Stab}_G(Y)$ , there is a natural homomorphism  $G^Y \rightarrow \text{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -, - \rangle)$ . Let  $\widetilde{G}^Y$  be the double cover of  $G^Y$  induced by the metaplectic double cover of  $\text{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -, - \rangle)$ .

**Definition 8.2.** The nilpotent  $\mathcal{O}$  is called admissible if the double cover  $\widetilde{G}^Y$  is split over the identity component of  $(G^Y)_0$ . It is called quasi-admissible if  $\widetilde{G}^Y$  has a finite-dimensional genuine representation.

Result of [GGS21] says that  $\text{WS}(\pi)$  consists of quasi-admissible orbits. But they do not know whether the non-admissible quasi-admissible orbits appear in Whittaker supports of representations. We may ask the same question for the wavefront set:

**Problem 1:** Are all the nilpotent orbits in the wavefront cycle admissible?

For the case when  $G$  is general linear, it is well known that the wavefront set  $\text{WF}(\pi)$  of an irreducible Casselman-Wallach representation  $\pi$  of  $G$  consists of a single nilpotent adjoint orbit. Together with some counting results, it maybe possible to construct all irreducible representations of general linear group via theta correspondence and coherent continuation.

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<sup>2</sup>This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre.

**Problem 2:** Can we construct all irreducible representations via theta correspondence, coherent continuation, and twist of characters?

Assuming the construction steps above are done, it then remains to study the Whittaker support  $\text{WS}(\pi)$  of  $\pi$ . We focus on two types of interactions with Whittaker models - theta correspondence and coherent continuation.

On one hand, Gomez-Zhu [GZ14] proved for type I reductive (symplectic-orthogonal) dual pairs in the stable range that the big theta lift preserves the structure of Whittaker models:

$$\text{Wh}_{\Theta(\mathcal{O})}(\Theta(\pi)) = \text{Wh}_{\mathcal{O}}(\pi^{\vee}).$$

Noting that  $(G, G') = (\text{GL}_n(F), \text{GL}_m(F))$  is a (reductive) dual pair for any  $n, m \in \mathbb{N}$ , one observes that if one can prove an analogue of Gomez-Zhu's result for type II dual pairs, one can use it to infer properties of the structure of the Whittaker support  $\text{WS}(\pi)$ . Hence we propose the following subconjecture:

**Conjecture 8.3.** *If  $(G, G') = (\text{GL}_n(F), \text{GL}_m(F))$  is a type II reductive dual pair over  $F$  in the stable range (with  $n \leq m$ ), and  $\pi$  is a genuine Casselman-Wallach representation of  $G$ , then*

$$\text{Wh}_{\Theta(\mathcal{O})}(\Theta(\pi)) = \text{Wh}_{\mathcal{O}}(\pi^{\vee}),$$

where  $\pi^{\vee}$  denotes the smooth contragredient of  $\pi$ . Furthermore, the above will imply that

$$\text{WS}(\Theta(\pi)) = \text{WS}(\pi^{\vee}).$$

On the other hand, Barbash-Vogan [BV80] proved that under coherent continuation, the multiplicity of wavefront cycle changes in the form of a Goldie rank polynomial, i.e.

$$\mathcal{WF}(\pi(\lambda)) = p(\lambda) \cdot \left( \sum_i c_i[\mathcal{O}_i] \right),$$

where  $p(\lambda)$  is the goldie rank polynomial,  $\lambda$  is dominant. We expect some similar result for the multiplicity of the generalized Whittaker model.

**Problem 3:** For a fixed nilpotent orbit  $\mathcal{O}$ , does there exist a constant  $c_{\mathcal{O}}$  such that  $\dim \text{Wh}_{\mathcal{O}}(\pi(\lambda)) = c_{\mathcal{O}} \cdot p(\lambda)$ ?

## 9. 2025.10.1

Orbit method for nilpotent Lie groups.

Let  $G$  be a connected, simply connected nilpotent Lie group with Lie algebra  $\mathfrak{g}$ . Let  $\mathfrak{g}^*$  be the dual space of  $\mathfrak{g}$ .

We have fourier transform:

$$\mathcal{F} : \mathcal{S}(\mathfrak{g})dX \rightarrow \mathcal{S}(\sqrt{-1}\mathfrak{g}^*), \quad f dX \mapsto \widehat{f dX}, \quad \widehat{f dX}(l) = \int_{\mathfrak{g}} f(X) e^{\langle l, X \rangle} dX.$$

Take its dual:

$$\mathcal{S}'(\sqrt{-1}\mathfrak{g}^*) \rightarrow (\mathcal{S}(\mathfrak{g})dX)', \quad T \mapsto \check{T}, \quad \langle \check{T}, f dX \rangle = \langle T, \widehat{f dX} \rangle.$$

For any irreducible unitary representation  $\pi$  of  $G$ , we have its character  $\Theta_\pi \in C^{-\infty}(G)$  defined by

$$\Theta_\pi(f) = \text{Tr}(\pi(f dg)), \quad f dg \in C_c^\infty(G) dg.$$

Using the exponential map  $\exp : \mathfrak{g} \rightarrow G$  (isomorphism in this case), we can view  $\Theta_\pi$  as a distribution on  $\mathfrak{g}$ :

$$\theta_\pi(f dX) = \Theta_\pi(f \circ \exp), \quad f dX \in C_c^\infty(\mathfrak{g}) dX.$$

In fact,  $\theta_\pi$  is a tempered generalized function on  $\mathfrak{g}$  (it can continuously extends to  $\mathcal{S}(\mathfrak{g})dX$ ). By the work of Howe, Kirillov, and others, we have the following theorem:

**Theorem 9.1.** *There exists a unique coadjoint orbit  $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$  such that  $\check{\theta}_\pi = \mu_{\mathcal{O}}$ , where  $\mu_{\mathcal{O}}$  is the canonical measure on  $\mathcal{O}$ .*

This defines a bijection between  $\widehat{G}$  (the unitary dual of  $G$ ) and the set of coadjoint orbits in  $\sqrt{-1}\mathfrak{g}^*$ .

Conversely, for any coadjoint orbit  $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$ , we can construct an irreducible unitary representation  $\pi_{\mathcal{O}}$  of  $G$  such that  $\check{\theta}_{\pi_{\mathcal{O}}} = \mu_{\mathcal{O}}$ . This is the more interesting part, use the method of geometric quantization.

Take any element  $\lambda \in \mathcal{O}$ , we have an identification  $\mathcal{O} = G/G_\lambda$ ,  $\lambda : \mathfrak{g}_\lambda \rightarrow \mathbb{C}$  is a Lie algebra homomorphism, and hence  $\chi_\lambda : G_\lambda \rightarrow \mathbb{C}^\times$  defined by  $\chi_\lambda(\exp X) = e^{\langle \lambda, X \rangle}$  is a unitary character of  $G_\lambda$ . Then we can defined a Hermitian line bundle  $\mathcal{L}_\lambda = G \times^{G_\lambda} \mathbb{C}$  over  $\mathcal{O}$ , with a natural  $G$ -invariant connection  $\nabla$  such that  $\text{curv}(\nabla) = \omega_{\mathcal{O}}$ , where  $\omega_{\mathcal{O}}$  is the Kirillov-Kostant-Souriau symplectic form on  $\mathcal{O}$ .

Then we can define the Hilbert space of square integrable sections of  $\mathcal{L}_\lambda$ , but this space is too large. We need to choose a polarization  $\mathfrak{h} \subseteq \mathfrak{g}_{\mathbb{C}}$  (maximal isotropic subalgebra with respect to the bilinear form  $(X, Y) \mapsto \langle \lambda, [X, Y] \rangle$ ) and consider the space of square integrable holomorphic sections of  $\mathcal{L}_\lambda$  which are covariant constant along  $\mathfrak{h}$ .

In fact the polarization  $\mathfrak{h}$  corresponds to a  $G$ -invariant integrable distribution on  $\mathcal{O}$ , its a subbundle  $P$  of  $T\mathcal{O}$  each fiber is a lagrangian subspace. Then we can define the space of polarized sections:

$$\Gamma_P(\mathcal{O}, \mathcal{L}_\lambda) = \{s \in \Gamma^\infty(\mathcal{O}, \mathcal{L}_\lambda \otimes |\omega_{\mathcal{O}/P}|^{\frac{1}{2}}) \mid \nabla_X s = 0, \forall X \in P\}.$$

On this space, we have a unitary  $G$  action and get a unitary representation  $\pi_{\mathcal{O}}$  of  $G$ , it turns out that  $\pi_{\mathcal{O}}$  is irreducible and independent of the choice of the polarization  $\mathfrak{h}$ .

Kirillov's philosophy works well for nilpotent Lie groups, but for more general Lie groups, it is not true anymore.

Now assume  $G$  is a real reductive group, with Lie algebra  $\mathfrak{g}_0$ . For any irreducible Casselman-Wallach representation  $\pi$  of  $G$ , we still have its generalized function character  $\Theta_\pi \in C^{-\infty}(G)$ , and hence a tempered generalized function  $\theta_\pi$  on  $\mathfrak{g}$ . Here  $\theta_\pi \in C^{-\infty}(\mathfrak{g}_0)$  is defined as follows:

$$\theta_\pi(f dX) = \Theta_\pi(\exp_*(\sqrt{\det(\exp)} f dX)), \quad f dX \in C_c^\infty(\mathfrak{g}_0) dX.$$

The generalized function  $\Theta_\pi$  has the following properties:

- $\Theta_\pi$  is invariant under conjugation of  $G$ ;
- $\Theta_\pi$  is an eigendistribution of  $\mathcal{U}(\mathfrak{g})^G$ ;
- $\Theta_\pi$  is locally integrable, and real analytic on the regular semisimple locus  $G^{\text{rs}}$ .

By the work of Rossmann, we have the following theorem:

**Theorem 9.2.** *If  $\pi$  is tempered, then*

$$\theta_\pi|_{\text{nbhd of } 0} = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}|_{\text{nbhd of } 0}.$$

Here  $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$ , is the orbital integral along  $\mathcal{O}_i$ .

For an arbitrary generalized function  $\Theta \in C^{-\infty}(M)$ , we can define its wavefront set  $\text{WF}(\Theta) \subseteq \sqrt{-1}\text{T}^*M$  as follows:

**Definition 9.3.** For any  $(x, \xi) \in \sqrt{-1}\text{T}^*M$ ,  $(x, \xi) \notin \text{WF}(\Theta)$  if there exists a compactly supported smooth function  $\varphi$  on  $M$  such that  $\varphi(x) \neq 0$ , and an open conic neighborhood  $\Gamma$  of  $\xi$  such that for any  $N \in \mathbb{N}$ , there exists a constant  $C_N > 0$  such that

$$|\widehat{\varphi\Theta}(t\eta)| \leq C_N t^{-N}, \quad \forall t > 0, \eta \in \Gamma.$$

$\text{WF}(\Theta)$  is a closed conic subset of  $\sqrt{-1}\text{T}^*M$  (singular direction of  $\Theta$ ).

In particular, we can define the wavefront set  $\text{WF}(\pi)$  of an irreducible Casselman-Wallach representation  $\pi$  of  $G$  as  $\text{WF}(\Theta_\pi) \cap (\sqrt{-1}\text{T}_1^*(G))$ , it is a closed conic  $G$ -invariant subset of  $\sqrt{-1}\mathfrak{g}_0^*$ . The GK-dimension of  $\pi$  is defined as

$$\text{Dim}(\pi) = \frac{1}{2} \dim(\text{WF}(\pi)).$$

$\theta_\pi$  is a real analytical function on  $\mathfrak{g}_0^{\text{rs}}$ , and

$$\theta_\pi^*(X) := \lim_{t \rightarrow 0} t^{\text{Dim}(\pi)} \theta_\pi(tX)$$

is still a real analytical function on  $\mathfrak{g}_0^{\text{rs}}$ , where  $\mathcal{B}$  is the flag variety of  $G$ . We can consider it as in  $C^{-\infty}(\mathfrak{g}_0)$ .

**Theorem 9.4.** (*Barbasch-Vogan*) *The asymptotic of  $\theta_\pi$  defined above is a non-negative linear combination of Fourier transforms of nilpotent orbital integrals:*

$$\theta_\pi^* = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}.$$

Here  $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$ , is the orbital integral along  $\mathcal{O}_i$ .

$$\text{WF}(\pi) = \bigcup_{c_i \neq 0} \overline{\mathcal{O}_i}.$$

## 10. 2025.10.2

**10.1. Conormal bundle of singular subvariety.** Let  $\mathcal{B}$  be the flag variety of  $G$ .  $\overline{\mathcal{B}}_y \subsetneq \overline{\mathcal{B}}_w$  two schubert varieties, what is the relation between their conormal bundles  $T_{\mathcal{B}_y}^* \mathcal{B}$  and  $T_{\mathcal{B}_w}^* \mathcal{B}$ ?

Let  $X$  be a smooth algebraic variety over  $\mathbb{C}$ ,  $Z \subseteq X$  be a closed subvariety, define the conormal bundle  $T_Z^* X$  of  $Z$  in  $X$  as follows:

$$T_Z^* X = \overline{T_{Z^{\text{reg}}}^* X}.$$

For a singular point  $z \in Z$  what is the fiber of  $T_Z^* X$  at  $z$ ?

Here is a simple example:

**Example 10.1.** Let  $X = \mathbb{C}^2$  and

$$Z = \{xy = 0\} \subset X$$

be the union of the coordinate axes.

- The regular locus is

$$Z^{\text{reg}} = (x\text{-axis} \setminus \{0\}) \cup (y\text{-axis} \setminus \{0\}).$$

- The singular locus is

$$S := Z \setminus Z^{\text{reg}} = \{0\}.$$

Now, the fibre of the conormal bundle at the origin is

$$(T_Z^* X)_0 = \text{Ann}(T_{x\text{-axis}}) \cup \text{Ann}(T_{y\text{-axis}}) = \langle dy \rangle \cup \langle dx \rangle,$$

which is a union of two lines in  $T_0^* \mathbb{C}^2$ .

On the other hand,

$$(T_S^* X)_0 = (T_{\{0\}}^* X)_0 = T_0^* X,$$

the entire cotangent plane.

Thus,

$$(T_Z^* X)_0 \subsetneq (T_{Z-Z^{\text{reg}}}^* X)_0,$$

**10.2. Summary of paper [KT84, The characteristic cycles of holonomic systems on a flag manifold].** Let  $G$  be a semisimple simply connected complex algebraic group. Define Steinberg variety:

$$Z = \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \mid x \in \text{Lie}(B') \cap \text{Lie}(B'')\} (= \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \mid \}).$$

(=  $\{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \mid x \in \mathfrak{n}'^\perp \cap \mathfrak{n}''^\perp\}$  more naturally.)

Here  $\mathcal{N} \subseteq \mathfrak{g}$  is the nilpotent cone of  $\mathfrak{g}$ ,  $\mathcal{B}$  is the flag variety of  $G$ . It can be identify with the union of conormal bundles of all  $G$ -orbits in  $\mathcal{B} \times \mathcal{B}$ :

$$Z = \bigcup_{w \in W} T_{\mathcal{Y}_w}^* (\mathcal{B} \times \mathcal{B}),$$

which is a closed conic lagrangian subvariety of  $T^*(\mathcal{B} \times \mathcal{B})$ .

We also have the moment map:

$$\mu : Z \rightarrow \mathcal{N}, \quad (x, B', B'') \mapsto x.$$

By taking characteristic cycles, we get a map from the (complex) Grothendieck group of  $G$ -equivariant holonomic  $D$ -modules on  $\mathcal{B} \times \mathcal{B}$  to the top Borel-Moore homology of  $Z$ :

$$\text{CC} : K_0(\text{Mod}_{\Delta G}^{\text{Coh}}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}})) \rightarrow H_{\text{top}}^{\text{BM}}(Z, \mathbb{C}).$$

We also have isomorphisms:

$$h : K_0(\text{Mod}_{\Delta G}^{\text{Coh}}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}})) \rightarrow \mathbb{C}[W], \quad \mathcal{M}_w \mapsto w, \quad \mathcal{L}_w \mapsto a(w)$$

By Kazhdan-Lusztig, we have:

$$a(w) = \sum_{y \leq w} (-1)^{l(y)+l(w)} P_{y,w}(1)y.$$

There is a convolution product on  $K_0(\text{Mod}_{\Delta G}^{\text{Coh}}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}}))$  which makes  $h$  an isomorphism of algebras.

On

$$H_{\text{top}}^{\text{BM}}(Z, \mathbb{C}) \simeq \bigoplus_{w \in W} H_{\text{top}}^{\text{BM}}(\overline{T_{\mathcal{Y}_w}^*(\mathcal{B} \times \mathcal{B})}, \mathbb{C}) \simeq \bigoplus_{w \in W} \mathbb{C}[\overline{T_{\mathcal{Y}_w}^*(\mathcal{B} \times \mathcal{B})}],$$

In the paper of Kazhdan-Lusztig [KL80], they use topological method to construct a  $W \times W$  action on  $H_{\text{top}}^{\text{BM}}(Z, \mathbb{C})$ .

Moreover there is a  $W \times W$  module decomposition:

$$H_{\text{top}}^{\text{BM}}(Z, \mathbb{C}) \simeq \bigoplus_{\mathcal{O}} H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}),$$

where  $\mathcal{O}$  runs over all nilpotent orbits in  $\mathfrak{g}^*$ ,  $Z_{\mathcal{O}} = \mu^{-1}(\mathcal{O})$ . There is also an isomorphism of  $W \times W$ -modules:

$$H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq (H_{\text{top}}^{\text{BM}}(\mathcal{B}_x, \mathbb{C}) \otimes H_{\text{top}}^{\text{BM}}(\mathcal{B}_x, \mathbb{C}))_{A_G(x)},$$

( $A_G(x)$  coinvariants) where  $x \in \mathcal{O}$ ,  $\mathcal{B}_x = \{B' \in \mathcal{B} \mid x \in \text{Lie}(B')\}$  (the Springer fiber),  $A_G(x) = \text{Stab}_G(x)/\text{Stab}_G(x)^0$ .

By classical Springer theory,  $H_{\text{top}}^{\text{BM}}(\mathcal{B}_x, \mathbb{C})$  is a representation of  $W \times A_G(x)$ , and we have the following decomposition:

$$H_{\text{top}}^{\text{BM}}(\mathcal{B}_x, \mathbb{C}) \simeq \bigoplus_{\rho \in \widehat{A_G(x)}} \tau(x, \rho) \otimes \rho.$$

Here  $\tau(x, \rho)$  is an irreducible representation of  $W$  or zero.

So we have a decomposition of  $W \times W$ -modules:

$$H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq \bigoplus_{\rho \in \widehat{A_G(x)}} \tau(x, \rho) \otimes \tau(x, \rho).$$

Combine the above together, we finally have the following decomposition:

$$H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq \bigoplus_{\rho \in \widehat{A_G(x)}} \tau(x, \rho) \otimes \tau(x, \rho), \quad H_{\text{top}}^{\text{BM}}(Z, \mathbb{C}) \simeq \bigoplus_{\mathcal{O}} H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}).$$

For any  $w \in W$ , there is a unique nilpotent orbit  $\text{St}(w)$  such that  $\mu(\overline{T_{\mathcal{Y}_w}^*(\mathcal{B} \times \mathcal{B})}) = \overline{\text{St}(w)}$ . The map  $w \mapsto \text{St}(w)$  defines a **surjective(?)** map from  $W$  to the set of nilpotent orbits in  $\mathfrak{g}^*$  (this map can be computed via Robinson-Schensted algorithm).

**Corollary 10.2.** *For any nilpotent orbit  $\mathcal{O}$ , there is an isomorphism of  $W \times W$ -modules:*

$$\bigoplus_{w \in W, \text{St}(w) = \mathcal{O}} \mathbb{C}[\overline{T_{\mathcal{Y}_w}^*(\mathcal{B} \times \mathcal{B})}] \simeq \bigoplus_{\rho \in \widehat{A_G(\mathcal{O})}} \tau(\mathcal{O}, \rho) \otimes \tau(\mathcal{O}, \rho).$$



**Theorem 10.3.** (*Main theorem of [KT84]*) *The map*

$$\mathrm{CC} : K_0(\mathrm{Mod}_{\Delta G}^{\mathrm{Coh}}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}})) \rightarrow H_{\mathrm{top}}^{\mathrm{BM}}(Z, \mathbb{C}),$$

*is an isomorphism and  $W \times W$ -equivariant.*

Define  $b(w) = h \circ \mathrm{CC}^{-1}(\overline{[T_{y_w}^* \mathcal{B} \times \mathcal{B}]})$ . Now we have three basis of  $\mathbb{C}[W]$ :

$$\{w\}_{w \in W}, \quad \{a(w)\}_{w \in W}, \quad \{b(w)\}_{w \in W}.$$

It was conjectured by Kashiwara and Tanisaki that:  $a(w) = b(w)$  when  $G = \mathrm{SL}_n(\mathbb{C})$  which means the characteristic cycle of  $\mathcal{L}_w$  is irreducible, but unfortunately it is not true. Counterexample exists when  $G = \mathrm{SL}_8(\mathbb{C})$ , where Kashiwara-Saito singularities appears in some Schubert varieties.

11. 2025.10.5

**The Springer representation**

Let  $G$  be a connected complex reductive group,  $K$  a symmetric subgroup of  $G$ . We can define the Springer representation of  $W$  on  $H_c^\bullet(T_K^*\mathcal{B}, \mathbb{Q})$ .

Recall that we have the following diagram:

$$\begin{array}{ccccc} \widetilde{\mathcal{N}} & \longrightarrow & \widetilde{\mathfrak{g}} & \longleftarrow & \widetilde{\mathfrak{g}}^{\text{rs}} \\ \downarrow p_{\mathcal{N}} & & \downarrow p & & \downarrow p_{\text{rs}} \\ \mathcal{N} & \longrightarrow & \mathfrak{g} & \longleftarrow & \mathfrak{g}^{\text{rs}} \end{array}$$

In this diagram  $p_{\text{rs}}$  is a  $W$ -torsor. So we have a natural  $W$ -action on  $p_{\text{rs},*}\mathbb{Q}_{\widetilde{\mathfrak{g}}^{\text{rs}}}$  and hence on  $\text{IC}(\mathfrak{g}^{\text{rs}}, \mathcal{L})$ , where  $\mathcal{L} = (p_{\text{rs}})_*\mathbb{Q}_{\widetilde{\mathfrak{g}}^{\text{rs}}}$ . There is a canonical isomorphism of perverse sheaves  $\text{IC}(\mathfrak{g}^{\text{rs}}, \mathcal{L}) \simeq \mathbb{R}p_*\mathbb{Q}_{\widetilde{\mathfrak{g}}^{\text{rs}}}[\dim(\mathfrak{g})]$ , so there is a natural  $W$ -action on  $\mathbb{R}p_*\mathbb{Q}_{\widetilde{\mathfrak{g}}^{\text{rs}}}$ . By proper base change, we have

$$\mathbb{R}p_{\mathcal{N},*}\mathbb{Q}_{\widetilde{\mathcal{N}}} \simeq \mathbb{R}p_*\mathbb{Q}_{\widetilde{\mathfrak{g}}} \big|_{\mathcal{N}}. \quad (\text{Spr} = \mathbb{R}p_{\mathcal{N},*}\mathbb{Q}_{\widetilde{\mathcal{N}}}[\dim \mathcal{N}] \text{ called } \textit{The Springer sheaf})$$

So there is a natural  $W$ -action on  $\mathbb{R}p_{\mathcal{N},*}\mathbb{Q}_{\widetilde{\mathcal{N}}}$  (actually  $\text{End}_{\text{per}(\mathcal{N})}(\text{Spr}) = \mathbb{Q}[W]$ ) and hence on  $H_c^\bullet(T_K^*\mathcal{B}, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\mathcal{N}, \mathbb{R}p_{\mathcal{N},*}\mathbb{Q}_{\widetilde{\mathcal{N}}})$  and  $H_c^\bullet(\mathcal{B}_x, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\{x\}, \mathbb{R}p_{\mathcal{N},*}\mathbb{Q}_{\widetilde{\mathcal{N}}})$ . (action on Springer fiber)

Since  $T_K^*\mathcal{B} = p_{\mathcal{N}}^{-1}(\mathfrak{p})$ , we also have a natural  $W$ -action on

$$H_c^\bullet(T_K^*\mathcal{B}, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\mathfrak{p} \cap \mathcal{N}, (\mathbb{R}p_{\mathcal{N},*}\mathbb{Q}_{\widetilde{\mathcal{N}}}) \big|_{\mathfrak{p} \cap \mathcal{N}}).$$

Finally, there is a  $W$ -action on  $(H_c^{2d}(T_K^*\mathcal{B}, \mathbb{Q}))^* = \bigoplus_{\mathcal{O} \in K \backslash \mathcal{B}} \mathbb{Q}[\overline{T_{\mathcal{O}}^*\mathcal{B}}]$ .

**Theorem 11.1** (Tanisaki). *The map:*

$$\text{CC} : K_0(\text{Mod}_K^{\text{Coh}}(\mathcal{D}_{\mathcal{B}})) \rightarrow (H_c^{2d}(T_K^*\mathcal{B}, \mathbb{Q}))^* = H_{2d}^{\text{BM}}(T_K^*\mathcal{B}, \mathbb{Q}),$$

*is surjective and  $W$ -equivariant.*

## 12. 2025.10.9

$X = \text{Spec}(R)$  smooth affine with coordinate  $x_1, \dots, x_n$ ,  $M$  a finitely generated  $\mathcal{D}_X$ -module.

**Definition 12.1.**  $M$  is called holonomic if  $\dim(\text{Ch}(M)) = n$ .

**Corollary 12.2.** *If  $M$  is holonomic, then  $\text{Ch}(M)$  is lagrangian in  $T^*X$ , and  $\text{CC}(M) = \sum_{E_i \subseteq X, \text{irr}} n_i [\overline{T_{E_i}^* X}]$ .*

The de-Rham complex and analytification:

$$\mathcal{K}(M; \partial_1) = [M \xrightarrow{\partial_1} M],$$

$$\mathcal{K}(M; \partial_1, \partial_2) = [M \xrightarrow{(\partial_1, \partial_2)} M \oplus M \xrightarrow{(\partial_2, -\partial_1)} M].$$

We can define  $\mathcal{K}(M; \partial_1, \dots, \partial_n)$  inductively.

There is another way to define the de-Rham complex, independent of the choice of coordinates:

$$\text{DR}(M) = [M \xrightarrow{\nabla} \Omega_X^1 \otimes M \xrightarrow{\nabla} \dots \xrightarrow{\nabla} \Omega_X^n \otimes M].$$

**Example 12.3.** Let  $X = \mathbb{C}$ ,  $M = \mathbb{C}[x, x^{-1}] \cdot x^\alpha$ ,  $\alpha \in \mathbb{C}$ ,  $\partial_x x^\alpha = \alpha \frac{1}{x} x^\alpha$ . Then  $M_\alpha = \mathcal{D}_X / \mathcal{D}_X(x\partial_x - (\alpha - k))$ . To get the *multiplicities* (the characteristic cycle), we examine the primary decomposition of the defining principal-symbol ideal:

$$(x\xi) = (x) \cap (\xi).$$

Each factor appears with exponent 1, and the associated graded module localized at a generic point of either component has length 1. Concretely:

- Localize at the generic point of  $V(\xi)$  (i.e. at the prime  $(\xi)$ ). In that localization,  $\xi$  is nonunitally small and  $x$  is a unit, so

$$(\text{gr } M_\alpha)_{(\xi)} \cong \mathbb{C}[x, \xi]_{(\xi)} / (\xi)$$

which is a copy of  $\mathbb{C}[x]_{(0)}$  of length 1 over the local ring — so the multiplicity is 1 on the zero section.

- Localize at the generic point of  $V(x)$  (i.e. at  $(x)$ ). There  $\xi$  is a unit and

$$(\text{gr } M_\alpha)_{(x)} \cong \mathbb{C}[x, \xi]_{(x)} / (x)$$

which is a copy of  $\mathbb{C}[\xi]_{\text{gen}}$  of length 1 — so the multiplicity is 1 on the conormal  $T_0^*X$ .

Hence both components appear with multiplicity 1.

Therefore the characteristic cycle is

$$\boxed{\text{CC}(M_\alpha) = [T_X^* X] + [T_0^* X].}$$

This calculation is independent of  $\alpha$ , because the parameter  $\alpha$  contributes only lower-order terms that do not affect the principal symbol.

Now consider the de-Rham complex of  $M_\alpha$ :

$$\text{DR}(M_\alpha) = [M_\alpha \xrightarrow{\partial_x} M_\alpha].$$

It is exact, this is bad. To fix this, we need to analytify  $M_\alpha$ :

$$M_\alpha^{\text{an}} = \mathcal{O}_{X^{\text{an}}} \otimes_{\mathbb{C}[x]} M_\alpha = \mathcal{O}_{X^{\text{an}}}[x^{-1}] \cdot x^\alpha.$$

**Example 12.4.**  $N = \mathbb{C}[\delta]$  where  $\delta$  is the delta function at 0,  $x \cdot \delta = 0$ . Then  $N = \mathcal{D}_X / \mathcal{D}_X x$ . Characteristic cycle is  $[T_0^* X]$  with multiplicity 1.

**Remark 12.5.** The two examples above consists of all regular singularities in some sence.

## 13. 2025.10.13

**Lemma 13.1.** *Let  $G$  be a connected Lie group,  $\tilde{G}$  is a double cover. Then the following are equivalent:*

- $\tilde{G}$  split;
- $\tilde{G}$  is disconnected;
- there exists a genuine representation of  $\tilde{G}$  which is trivial on the identity component  $\tilde{G}_0$ .

*Proof.* Only need to notice that  $\tilde{G}_0 \rightarrow G$  is a covering map of degree 1 or 2.  $\square$

There is a same statement for p-adic groups in [Nev99].

**Lemma 13.2.** *Let  $G$  be an arbitrary Lie group,  $\pi : \tilde{G} \rightarrow G$  a double cover. Then the following are equivalent:*

- there exists a genuine finite dimensional representation of  $\tilde{G}$  which is trivial on the identity component  $\tilde{G}_0$ ;
- $z \notin \tilde{G}_0$ ;
- $z \notin (\pi^{-1}(G_0))_0$ ;
- there exist a genuine finite dimensional representation of  $\pi^{-1}(G_0)$  which is trivial on the identity component  $(\pi^{-1}(G_0))_0$ ;
- $\pi^{-1}(G_0)$  is disconnected;
- $\pi^{-1}(G_0)$  is split.

Here  $z$  is the nontrivial element in  $\ker(\pi)$ .

If in addition the double cover  $\tilde{G}$  is induced by a square root of a character  $\gamma : G \rightarrow \mathbb{C}^\times$ , i.e.  $\tilde{G} = \{(g, \epsilon) \in G \times \mathbb{C}^\times \mid \epsilon^2 = \gamma(g)\}$ , these are also equivalent to:

- $\gamma$  has a square root.

*Proof.* Only need to notice that  $(\pi^{-1}(G_0))_0 = \tilde{G}_0$ , together with results in [Sch87].  $\square$

14. 2025.10.14

$X = \text{Spec}(R)$  smooth affine with coordinate  $x_1, \dots, x_n$ ,  $M$  is an integrable connection.

**Theorem 14.1.**  $M^{an} = L \otimes_{\mathbb{C}} \mathcal{O}_{X^{an}}$  for some local system  $L$  on  $X^{an}$ .

*Proof.* We know that  $M$  is locally free, suppose  $M|_U$  is freely generated by  $m_1, \dots, m_n$ , suppose  $\partial_n \underline{m} = \chi \underline{m}$  where  $\chi$  is a matrix of holomorphic functions on  $U$ . Locally we can expand  $\chi$  with respect to  $x_n$ :

$$\chi = \sum_{i=0}^{\infty} \chi_i(x_1, \dots, x_{n-1}) x_n^i.$$

Take  $\Psi = \sum_{i=1}^{\infty} \frac{1}{i+1} \chi_i(x_1, \dots, x_{n-1}) x_n^{i+1}$ , so  $\partial_n \Psi = \chi$ . Take  $\Phi = \exp(-\Psi)$  which is invertible, and set  $\underline{s} = \Phi \underline{m}$ .

So

$$\partial_n s = \partial_n(\Phi) \cdot \underline{m} + \Phi \cdot (\partial_n \underline{m}) = 0.$$

There is a small open subset  $V \subset U$  such that  $M^{an}|_V = \bigoplus_{i=1}^r \mathcal{O}_V s_i$ , take

$$\begin{aligned} M_1 &= \ker(M^{an}|_V \xrightarrow{\partial_n} M^{an}|_V) \\ &= \left\{ \sum_{i=1}^r f_i s_i \in M^{an}|_V \mid \partial_n \left( \sum_{i=1}^r f_i s_i \right) = 0, \forall i \right\} \\ &= \left\{ \sum_{i=1}^r f_i s_i \in M^{an}|_V \mid \partial_n f_i = 0, \forall i \right\} \\ &= \bigoplus_{i=1}^r \mathcal{O}_{V_1} s_i, \quad V_1 = V \cap \{x_n = 0\}. \end{aligned}$$

Continue this process, we can finally get a subsheaf

$$M_n|_V = \bigoplus_{i=1}^n \mathbb{C} s_i.$$

of  $M^{an}$ , where  $s_i$  are horizontal sections. So  $M^{an} = L \otimes_{\mathbb{C}} \mathcal{O}_{X^{an}}$  where  $L$  is the local system generated by  $s_i$ . □

**Remark 14.2.** The proof actually implies that integrable connections are locally generated by horizontal sections, and the sheaf of horizontal sections is a local system.

**Corollary 14.3.**  $\text{DR}(M) = L$

*Proof.*  $H^0(\text{DR}(M)) \simeq \ker(M \xrightarrow{\nabla} \Omega_X^1 \otimes M) = L$ . All the other cohomologies vanish because of Poincaré lemma, since this complex is locally de-Rham complex. □

**Remark 14.4.** This means that de Rham complex of  $M$  gives a locally free resolution of  $L$ . So the hypercohomology of  $\text{DR}(M)$  computes the cohomology of  $L$ .

15. 2025.10.16

**Kashiwara's equivalence**

$Y$  smooth affine  $/\mathbb{C}$  with coordinate  $x_1, \dots, x_n, y_1, \dots, y_l$ ,  $X = \{y_1 = \dots = y_l = 0\} \subset Y$ ,  $X$  is also smooth.  $I = (y_1, \dots, y_l)$ ,  $\mathcal{D}_Y$  the ring of differential operators on  $Y$ ,  $\mathcal{D}_X$  the ring of differential operators on  $X$ .  $\text{Mod}_{qc}(\mathcal{D}_X)$ : the category of quasi-coherent  $\mathcal{D}_X$  modules,  $\text{Mod}_{qc}^X(\mathcal{D}_Y)$ : the category of quasi-coherent  $\mathcal{D}_Y$  modules supported on  $X$ .

**Theorem 15.1.** (*Kashiwara's equivalence*) *The functor:*

$$i_+ : \text{Mod}_{qc}(\mathcal{D}_X) \rightarrow \text{Mod}_{qc}^X(\mathcal{D}_Y), \quad M \mapsto \mathcal{D}_{Y \leftarrow X} \otimes_{\mathcal{D}_X} M = M[\partial_{y_1}, \dots, \partial_{y_l}],$$

*is an equivalence of categories. Its quasi-inverse is given by:*

$$i^\# : \text{Mod}_{qc}^X(\mathcal{D}_Y) \rightarrow \text{Mod}_{qc}(\mathcal{D}_X), \quad N \mapsto \ker(N \xrightarrow{y} N).$$

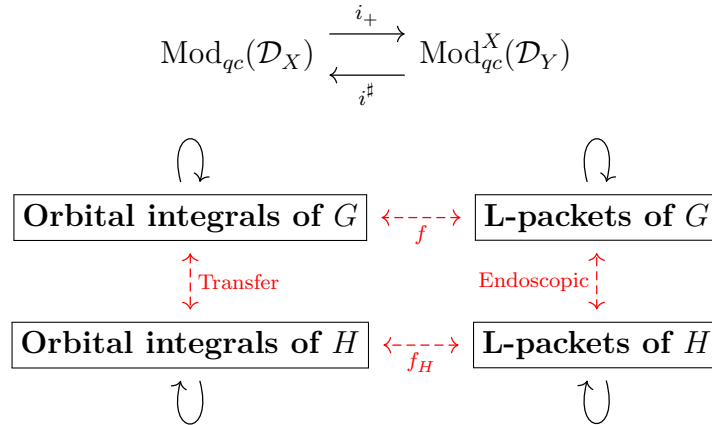
*Proof.* There is a proof in [HTT08], I don't want to repeat it here.  $\square$

**Remark 15.2.** In the case of quasi-coherent sheaves, the equivalence does not hold. For example, Let  $Y = \text{Spec}(R)$ ,  $X = \text{Spec}(R/I) \subseteq Y$  a closed subscheme defined by ideal  $I$ . Then the functor:

$$i^* : \text{Mod}_{qc}^X(\mathcal{O}_Y) \rightarrow \text{Mod}_{qc}(\mathcal{O}_X), \quad N \mapsto R/I \otimes_R N.$$

is not an equivalence of categories. Since if  $M$  is an  $R$ -module, then  $M/I^2M$  is supported on  $X$ , but it does not defined over  $R/I$ .

**Corollary 15.3.** *Let  $M$  be a quasi-coherent  $\mathcal{D}_X$ -module, if  $Ch(M) = T_Y^*X$  for an irreducible closed subvariety  $X \subset Y$ , then  $M = i_+(N)$  for some integrable connection  $N$  on  $Y$ .*



16. 2025.10.19

**Degenerate Whittaker model**

$G$  be a finite extension of a reductive algebraic group defined over local field  $F$ ,  $\mathfrak{g}$  be its Lie algebra, fix a nontrivial character  $\chi : F \rightarrow \mathbb{C}^\times$ .

A Whittaker pair is a pair  $(S, \varphi)$  where  $S \in \mathfrak{g}$  is a rational semi-simple element,  $\varphi \in \mathfrak{g}^*$  such that  $\text{ad}^*(S)(\varphi) = -2\varphi$ . Given such a Whittaker pair, we can define a degenerate Whittaker model  $\mathcal{W}_{S, \varphi}$  as follows: Let  $\mathfrak{g} = \bigoplus_{r \in \mathbb{Q}} \mathfrak{g}_r^S$  be the eigenspace decomposition with respect to  $\text{ad}(S)$ , define

$$\mathfrak{u} = \mathfrak{g}_{\geq 1} = \bigoplus_{r \geq 1} \mathfrak{g}_r^S.$$

Define an anti-symmetric form  $\omega_\varphi$  on  $\mathfrak{u}$  by

$$\omega_\varphi(X, Y) = \varphi([X, Y]), \quad X, Y \in \mathfrak{u}.$$

Let  $\mathfrak{n}$  be the kernel of  $\omega_\varphi$ , and  $\mathfrak{n}' = \mathfrak{n} \cap \ker(\varphi)$ . Then  $\mathfrak{u}/\mathfrak{n}$  is a symplectic space with symplectic form induced by  $\omega_\varphi$ , and  $\mathfrak{u}/\mathfrak{n}'$  is a Heisenberg Lie algebra with center  $\mathfrak{n}/\mathfrak{n}'$ . Corresponding groups are defined by exponentiation:

$$U = \exp(\mathfrak{u}), \quad N = \exp(\mathfrak{n}), \quad N' = \exp(\mathfrak{n}').$$

If  $\varphi \neq 0$ , there is a unique irreducible representation  $(\rho_\varphi, \mathcal{S}_\varphi)$  of  $U/N'$  with central character  $\chi_\varphi : N/N' \rightarrow \mathbb{C}^\times$  defined by

$$\chi_\varphi(\exp(X)) = \chi(\varphi(X)), \quad X \in \mathfrak{n}.$$

Define the degenerate Whittaker model:

$$\mathcal{W}_{S, \varphi} = \text{Ind}_U^G(\rho_\varphi).$$

If  $\varphi = 0$ , define  $\mathcal{W}_{S, 0} = \text{Ind}_U^G(1_U)$ .

**Remark 16.1.** In the special case when  $(S, \varphi)$  comes from an  $\mathfrak{sl}_2$ -triple  $(e, h, f)$  with  $S = h$  and  $\varphi$  corresponding to  $f$  via the Killing form, we can see that  $\mathfrak{n} = \mathfrak{g}_{\geq 2}$ . Then  $\mathcal{W}_f = \mathcal{W}_{S, \varphi}$  is called a generalized Whittaker model associated to the nilpotent orbit of  $f$ .

**Affine transformation group of real line**

$P_2(\mathbb{R}) = \mathbb{R}^\times \ltimes \mathbb{R}$ , with multiplication:

$$(a, b) \cdot (a', b') = (aa', b + ab').$$

Its Lie algebra is  $\mathfrak{p}_2(\mathbb{R}) = \mathbb{R} \times \mathbb{R}$  with bracket:

$$[(x_1, y_1), (x_2, y_2)] = (0, x_1 y_2 - x_2 y_1).$$



17. 2025.10.21

**Spectral Sequences of Filtrated Complexes**

$(F^\bullet, d)$  be a filtered complex, i.e. a complex  $F^\bullet$  with an increasing filtration of subcomplexes:

$$\cdots \subseteq F_j^\bullet \subseteq F_{j+1}^\bullet \subseteq \cdots \subseteq F^\bullet.$$

Such that  $d(F_j^i) \subseteq F_{j+1}^{i+1}$ , and the filtration is exhaustive and separated:

$$\bigcup_{j \in \mathbb{Z}} F_j^i = F^i, \quad \bigcap_{j \in \mathbb{Z}} F_j^i = 0.$$

For any triple  $(i, j, k)$ , define:

- $Z_j^i(k) := \{u \in F_j^i \mid du \in F_{j-k}^{i+1}\};$
- $B_j^i(k) := d(F_{j+k-1}^{i-1}) \cap F_j^i;$
- $E_j^i(k) := \frac{Z_j^i(k) + F_{j-1}^i}{B_j^i(k) + F_{j-1}^i}.$

So  $d(Z_j^i(k)) \subseteq Z_{j-k}^{i+1}(k)$ , which induces a differential  $d_k : E_j^i(k) \rightarrow E_{j-k}^{i+1}(k)$ .

We also define  $E_k^{p,q} := E_{-p}^{p+q}(k)$ .

**Example 17.1.**  $E_j^i(0) = \frac{F_j^i}{F_{j-1}^i}$ , so  $E_0^{p,q} = \text{Gr}_{-p} F^{p+q}$ .

$E_j^i(1) = H^i(\text{Gr}_j F^\bullet)$ , so  $E_1^{p,q} = H^{p+q}(\text{Gr}_{-p} F^\bullet)$ . The first page of the spectral sequence is just the cohomology of the associated graded complex.

**Lemma 17.2.**  $H^i(E_j^\bullet(k)) = E_j^i(k+1)$ .

**Definition 17.3.** The spectral sequence associated to the filtered complex  $(F^\bullet, d)$  is defined to be the collection of complexes  $\{(E_\bullet^i(k), d_k)\}_{k \geq 0}$ .

They convergent to the cohomology of the original complex:  $E_\infty^{p,q} \simeq \text{Gr}_{-p} H^{p+q}(F^\bullet)$ , if for any  $i, j$ , there exist  $k \gg 0$  such that  $E_j^i(k) = \text{Gr}_j H^i(F^\bullet)$ .

**Definition 17.4.** If there exist a  $w \in \mathbb{Z}_{\geq 0}$  such that for any  $i, j$ ,  $F_j^i \cap d(F_{j-w}^{i-1}) \subseteq d(F_{j+w}^{i-1})$ . Then the filtration is called  $w$ -regular.

**Theorem 17.5.** If  $F^\bullet$  is  $w$ -regular, then the spectral sequence associated to  $(F^\bullet, d)$  converges to the cohomology of the original complex at page  $w+1$ , i.e. for any  $i, j$ ,  $E_j^i(w+1) \simeq \text{Gr}_j H^i(F^\bullet)$ .

**Example 17.6.**  $K^\bullet$  is a chain complex,  $F$  a left exact functor, then there is a natural spectral sequence:

$$E_2^{p,q} = \mathbb{R}^{p+q} F(K^q) \Rightarrow \mathbb{R}^{p+q} F(K^\bullet).$$

(maybe:

$$E_1^{p,q} = \mathbb{R}^p F(K^q) \Rightarrow \mathbb{R}^{p+q} F(K^\bullet)?$$

)

Because we can use the Cartan-Eilenberg resolution to resolve  $K^\bullet$ , which is given by:

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & K^0 & \longrightarrow & I^{0,0} & \longrightarrow & I^{0,1} \longrightarrow \dots \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & K^1 & \longrightarrow & I^{1,0} & \longrightarrow & I^{1,1} \longrightarrow \dots \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & K^2 & \longrightarrow & I^{2,0} & \longrightarrow & I^{2,1} \longrightarrow \dots \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

Then we can form a total complex  $\text{Tot}(I^{\bullet,\bullet})$  with a natural filtration, and apply the functor  $F$  to get a filtered complex  $(F(\text{Tot}(I^{\bullet,\bullet})), d)$ . The spectral sequence associated to this filtered complex is just the spectral sequence above.

A special case is the Hodge spectral sequence, when  $X$  is a smooth algebraic variety over  $\mathbb{C}$ ,  $K^\bullet = \Omega_X^\bullet$  the de-Rham complex, and  $F = \mathbb{R}\Gamma(X, -)$  the derived global section functor, the spectral sequence is:

$$E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow H_{\text{dR}}^{p+q}(X).$$

18. 2025.10.23

**Lemma 18.1.** *Let  $H$  be a Lie group acting freely and properly on a smooth manifold  $M$ , then for any smooth manifold  $T$ , there is a natural isomorphism:*

$$\mathrm{Hom}_{sm}(T, M/H) \simeq \frac{\{(P \rightarrow^\pi T, f : P \rightarrow M) | \pi \text{ a principle } H\text{-bundle, } f \text{ is } H\text{-equivariant}\}}{\text{equivalence}}.$$

*Proof.* Given a smooth map  $g : T \rightarrow M/H$ , we can form the following Cartesian diagram:

$$\begin{array}{ccc} P & \xrightarrow{f} & M \\ \pi \downarrow & & \downarrow \\ T & \xrightarrow{g} & M/H \end{array}$$

Since  $H$  acts freely and properly on  $M$ ,  $\pi : P \rightarrow T$  is a principle  $H$ -bundle, and  $f : P \rightarrow M$  is  $H$ -equivariant.

Conversely, given a principle  $H$ -bundle  $\pi : P \rightarrow T$  and an  $H$ -equivariant map  $f : P \rightarrow M$ , we can define a smooth map  $g : T \rightarrow M/H$  by sending  $t \in T$  to the orbit of  $f(p)$  where  $p \in \pi^{-1}(t)$ .  $\square$

**Remark 18.2.** Algebraic geometers use this to define the quotient stack  $[M/H]$  even if the action is not free or proper, but  $\mathrm{Hom}(T, [M/H])$  need to be a groupoid, not just a set of equivalence classes.

19. 2025.10.29

**Unipotent Arthur packets and special unipotent representation [ABV92]**

Let  $G$  be a connected reductive algebraic group over  $\mathbb{R}$ , fix a weak  $E$ -group  $\check{G}^\Gamma$ . Which is an extension:

$$1 \rightarrow \check{G} \rightarrow \check{G}^\Gamma \rightarrow \text{Gal}(\mathbb{C}/\mathbb{R}) \rightarrow 1.$$

Here  $\check{G}$  is the complex dual group of  $G$ . An  $L$ -parameter is a continuous homomorphism:

$$\phi : W_{\mathbb{R}} \rightarrow \check{G}^\Gamma,$$

such that the composition  $W_{\mathbb{R}} \rightarrow {}^L\check{G} \rightarrow \text{Gal}(\mathbb{C}/\mathbb{R})$  is the natural projection and  $\phi(\mathbb{C}^*)$  consists of semisimple elements of  $\check{G}$ .

Given an  $L$ -parameter  $\phi$ , we can attach the following data:

- two semisimple elements  $\lambda, \mu \in \check{G}$ , defined by  $\phi(e^t) = \exp(\lambda t + \mu \bar{t})$ ;
- $y = \exp(\pi i \lambda) \phi(j) \in \check{G}^\Gamma - \check{G}$ .

The  $L$ -parameter  $\phi$  is called *tempered* if the closure of  $\phi(W_{\mathbb{R}})$  in the complex analytic topology is compact. This is equivalent to say that  $\lambda + \text{Ad}(y)\lambda$  is an elliptic element of  $\check{\mathfrak{g}}$ .

Fix a geometric parameter  $(y, \Lambda)$ , it lies in a variety  $X_y(\mathcal{O}, \check{G}^\Gamma) \simeq \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))$ . The  $\check{G}$ -equivariant microlocal geometry on  $X_y(\mathcal{O}, \check{G}^\Gamma)$  is in some sense equivalent to the  $K(y)$ -equivariant microlocal geometry on  $\check{G}(\Lambda)_0/P(\Lambda)$ .

Roughly speaking, there is a bijection between:

$$\begin{array}{c} \boxed{\text{representation with minimal GK-dim}} \\ \updownarrow \\ \boxed{\text{parameters supported on regular orbits in } \check{G}(\Lambda)_0/P(\Lambda)} \end{array}$$

**Definition 19.1.** An Arthur parameter is a homomorphism:

$$\psi : W_{\mathbb{R}} \times SL(2, \mathbb{C}) \rightarrow \check{G}^\Gamma,$$

such that the restriction to  $W_{\mathbb{R}}$  ( $\psi_0$ ) is a tempered  $L$ -parameter, and the restriction to  $SL(2, \mathbb{C})$  ( $\psi_1$ ) is algebraic.

For an arthur parameter  $\psi$ , we can define an  $L$ -parameter  $\phi_\psi$  by:

$$\phi_\psi(w) = \psi(w, \begin{pmatrix} |w|^{1/2} & 0 \\ 0 & |w|^{-1/2} \end{pmatrix}).$$

Let  $y_1 = \psi_1 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \in \check{G}$ , and  $\lambda_1 = d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \in \check{\mathfrak{g}}$ ,  $(y_0, \lambda_0)$  is the parameter defined by the restriction of  $\psi$  to  $W_{\mathbb{R}}$ .

Then we have  $y = y_0 y_1$ ,  $\lambda = \lambda_0 + \lambda_1$ , since  $\phi_\psi(e^t) = \exp(\lambda_0 t + \mu_0 \bar{t}) \psi_1 \begin{pmatrix} |e^t| & 0 \\ 0 & |e^{-t}| \end{pmatrix}$ , and

$$\psi_1 \begin{pmatrix} |e^t| & 0 \\ 0 & |e^{-t}| \end{pmatrix} = \exp \left( t \cdot d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} + \bar{t} \cdot d\psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \right) = \exp(\lambda_1 t + \lambda_1 \bar{t}).$$

So  $\phi_\psi(e^t) = \exp((\lambda_0 + \lambda_1)t + (\mu_0 + \lambda_1)\bar{t})$ , hence  $\lambda = \lambda_0 + \lambda_1$ .

And

$$y = \exp(\pi i \lambda) \phi_\psi(j) = \exp(\pi i \lambda_0) \exp(\pi i d \psi_1 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}) \psi_0(j) = \exp(\pi i \lambda_0) \psi_1 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \psi_0(j) = y_0 y_1.$$

In addition of the two semisimple elements  $\lambda, \mu$  and  $y$  defined as above, we can also define a nilpotent element  $E_\psi \in \mathfrak{g}$  by:

$$E_\psi = d\psi_1 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{g}.$$

$E_\psi$  enjoys the following properties:

- $\text{Ad}(y)E_\psi = -E_\psi$ , this means  $E_\psi \in \mathfrak{s}(y)$ ;
- $[\lambda, E_\psi] = E_\psi$ , in other words,  $E_\psi \in \mathfrak{g}(\Lambda)_1 \subseteq \mathfrak{n}(\lambda)$  where  $\mathfrak{g}(\Lambda) = \bigoplus_{r \in \mathbb{Z}} \mathfrak{g}_r$  is the eigenspace decomposition with respect to  $\text{ad}(\lambda)$ .

Langlands parameter  $(y, \lambda)$



$$S = \check{G} \times^{K(y)} eP(\Lambda) \subseteq X_y(\mathcal{O}, \check{G}^\Gamma) = \check{G} \times^{K(y)} \check{G}(\Lambda)/P(\Lambda)$$



$$S_K = K(y) \cdot (eP(\Lambda)) \subseteq \check{G}(\Lambda)_0/P(\Lambda)$$

Cotangent space at the base point:

$$T_{eP(\Lambda)}^*(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \mathfrak{n}(\lambda).$$

Conormal space at the base point of the  $K(y)$ -orbit:

$$T_{K(y), eP(\Lambda)}^*(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \mathfrak{s}(y) \cap \mathfrak{n}(\lambda).$$

So we can view  $E_\psi$  as an element in the conormal space  $T_{K(y), eP(\Lambda)}^*(\check{G}(\Lambda)_0/P(\Lambda))$  also in  $T_{\check{G}, \phi_\psi}^*(X(\check{G}^\Gamma))$ .

**Definition 19.2.** An Arthur parameter  $\psi$  is called *unipotent* if the restriction to  $\mathbb{C}^* \subseteq W_{\mathbb{R}}$  is trivial.

**Remark 19.3.** In this case,  $\psi : \text{SL}_2(\mathbb{C}) \times \Gamma \rightarrow \check{G}^\Gamma$ . So  $\lambda = \lambda_1$

**Lemma 19.4.** Suppose  $\psi$  is unipotent. Consider the partial Springer resolution

$$\mu : T^*(\check{G}(\Lambda)_0/P(\Lambda)) \simeq \check{G}(\Lambda)_0 \times^{P(\Lambda)} \mathfrak{n}(\lambda) \rightarrow \mathcal{N}_{P(\Lambda)},$$

$E_\psi \in \mathcal{N}_{P(\Lambda)} \cap \mathfrak{s}(y)$  is a regular element (belong to the Richardson orbit), and the fiber  $\mu^{-1}(E_\psi)$  is a single point. This guarantees that  $\mu$  has degree 1.

*Proof.* We can see that

$$\mathfrak{p}(\Lambda) = \mathfrak{g}^{\text{Ad}(E_\psi)} + \mathfrak{g}^{\text{Ad}(E_\psi)^2} \cap \text{Ad}(E_\psi)\mathfrak{g} + \mathfrak{g}^{\text{Ad}(E_\psi)^3} \cap \text{Ad}(E_\psi)^2\mathfrak{g} + \cdots,$$

So the centralizer  $\check{G}(\Lambda)^{E_\psi} \subseteq P(\Lambda)$ , which means that the fiber  $\mu^{-1}(E_\psi)$  is a single point. The operator  $E_\psi$  carry the sum of non-negative eigenspaces of  $\text{ad}(\lambda)$  onto the sum of positive

eigenspaces, i.e.  $\text{Ad}(E_\psi)\mathfrak{p}(\Lambda) = \mathfrak{n}(\Lambda)$ . In other words, the differential of the map:

$$P(\Lambda) \rightarrow \mathfrak{n}(\Lambda), \quad p \mapsto \text{Ad}(p)E_\psi,$$

is surjective at the identity coset, so the orbit  $\check{G}(\Lambda) \cdot E_\psi$  is dense in  $\mathfrak{n}(\Lambda)$ , which means that  $E_\psi$  is a regular element in  $\mathcal{N}_{\mathcal{P}(\Lambda)}$ .  $\square$

**Theorem 19.5.** *Fix an algebraic morphism  $\psi : \text{SL}_2(\mathbb{C}) \rightarrow \check{G}^\Gamma$ , let  $\lambda = d\psi \left( \begin{smallmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{smallmatrix} \right)$ ,  $\mathcal{O} = \check{G} \cdot \lambda \subseteq \check{\mathfrak{g}}$  the corresponding semisimple orbits. Consider the open and closed subvariety of the geometric parameter space  $X_y(\mathcal{O}, \check{G}^\Gamma) \simeq \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))$ .*

*Then there are bijections between:*

- *equivalence classes of unipotent Arthur parameters supported on  $X_y(\mathcal{O}, \check{G}^\Gamma)$ ;*
- *regular  $K(y)$ -orbits in  $\check{G}(\Lambda)_0/P(\Lambda)$ ;*
- *regular  $K(y)$ -orbits in  $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$ .*

*Proof.* Given a unipotent Arthur parameter  $\psi$ , we can associate to it a nilpotent element  $E_\psi \in \mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$  as above, then  $K(y) \cdot E_\psi$  is a regular orbit in  $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$ .

Conversely, given a regular  $K(y)$ -orbit in  $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$ , let  $e$  be a representative. We can construct a homomorphism  $\psi'_1 : \text{SL}_2(\mathbb{C}) \rightarrow \check{G}$  such that  $d\psi'_1 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = e$ , and

$$\psi'_1(\theta_{\text{SL}_2}(x)) = \theta_y(\psi'_1(x)).$$

Define  $y_0 = y\psi'_1 \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$

Then  $(y_0, \psi'_1)$  is the corresponding unipotent Arthur parameter.  $\square$

**Remark 19.6.** In other words, this theorem tell us the following diagram:

$$\begin{array}{ccc} \boxed{\check{G}\text{-orbits in } \check{G} \times^{K(y)} (\check{G}(\Lambda)_0/P(\Lambda))} & \xleftarrow{1-1} & \boxed{K(y)\text{-orbits in } \check{G}(\Lambda)_0/P(\Lambda)} \\ \uparrow & & \uparrow \\ \boxed{\text{unipotent A-parameters supported on } X_y(\mathcal{O}, \check{G}^\Gamma)} & \xleftarrow{1-1} & \boxed{\text{regular } K(y)\text{-orbits in } \check{G}(\Lambda)_0/P(\Lambda)} \end{array}$$

**Theorem 19.7.** *Under the above setting, fix a unipotent Arthur parameter  $\psi$  supported on  $X_y(\mathcal{O}, \check{G}^\Gamma)$ , let  $S$  be the corresponding  $\check{G}$ -orbit in  $X_y(\mathcal{O}, \check{G}^\Gamma)$ ,  $S_K$  be the corresponding  $K(y)$ -orbit in  $\check{G}(\Lambda)_0/P(\Lambda)$ , and  $\mathcal{O}_K$  be the corresponding  $K(y)$ -orbit in  $\mathfrak{s}(y) \cap \mathcal{N}_{\mathcal{P}(\Lambda)}$ .*

*Then there are bijections between:*

- *irreducible  $(\check{\mathfrak{g}}(\Lambda), K(y)^{\text{alg}})$ -modules  $M$  annihilated by  $I_{\mathcal{P}(\Lambda)}$  with  $\overline{\mathcal{O}_K} \subseteq \text{AV}(M)$ ;*
- *irreducible  $K(y)^{\text{alg}}$ -equivariant  $\mathcal{D}$ -modules  $\mathcal{M}$  on  $\check{G}(\Lambda)_0/P(\Lambda)$  with  $T_{S_K}^*(\check{G}(\Lambda)_0/P(\Lambda)) \subseteq \text{Ch}(\mathcal{M})$ ;*
- *irreducible  $\check{G}$ -equivariant  $\mathcal{D}$ -modules  $\mathcal{N}$  on  $X_y(\mathcal{O}, \check{G}^\Gamma)$  with  $T_S^*(X_y(\mathcal{O}, \check{G}^\Gamma)) \subseteq \text{Ch}(\mathcal{N})$ ;*
- *Arthur packets  $\Pi_\psi^{\text{Art}}(G/\mathbb{R})$  associated to  $\psi$ .*

**Corollary 19.8.** *Fix a nilpotent orbit  $\check{\mathcal{O}} \subseteq \check{\mathfrak{g}}$ , attached to it is a semisimple orbit  $\check{\mathcal{O}}_{ss} \subseteq \check{\mathfrak{g}}$  then*

$$\mathrm{Unip}_{\check{\mathcal{O}}}(G/\mathbb{R}) = \bigcup_{\psi \text{ unipotent}, \mathcal{O}_{\psi} = \check{\mathcal{O}}_{ss}} \Pi_{\psi}^{Art}(G/\mathbb{R}).$$

**Remark 19.9.**

$$\mathrm{Unip}_{\check{\mathcal{O}}}(G/\mathbb{R}) = \{\text{irreducible representations in } \mathrm{Irr}_{\check{\mathcal{O}}_{ss}}(G/\mathbb{R}) \text{ with minimal GK-dim}\}$$

**Example 19.10.**  $G = G_1 \times G_1$  where  $G_1$  is a complex connected reductive algebraic group viewed as a real group. Then  $\check{G}^{\Gamma} = (\check{G}_1 \times \check{G}_1) \rtimes \{1, \delta\}$ . A homomorphism  $\psi : \mathrm{SL}_2(\mathbb{C}) \rightarrow \check{G}$  is determined by two homomorphisms  $\psi_1, \psi_2 : \mathrm{SL}_2(\mathbb{C}) \rightarrow \check{G}_1$ , we can check that  $\psi$  comes from a unipotent Arthur parameter if and only if  $\psi_1$  and  $\psi_2$  are conjugate to each other. So unipotent Arthur parameters of  $G$  are in bijection with homomorphisms  $\psi_1 : \mathrm{SL}_2(\mathbb{C}) \rightarrow \check{G}_1$ . Therefore, equivalence classes of unipotent Arthur parameters of  $G_1$  are in bijection with nilpotent orbits in  $\check{\mathfrak{g}}_1$ .

20. 2025.11.4

**Some questions about unitary dual problem**

- How can we reduce the classification of irreducible unitary  $(\mathfrak{g}, K)$ -module to the classification of such module with real infinitesimal character?
- Why there always exist a  $\mathfrak{u}_{\mathbb{R}}$ -invariant Hermitian form on irreducible  $(\mathfrak{g}, K)$ -module with real infinitesimal character?
- what is Jantzen filtration of a standard  $(\mathfrak{g}, K)$ -module? Why it coincides with Hodge filtration?

Let  $G$  be a connected reductive group over  $\mathbb{C}$ ,  $\mathcal{B}$  the variety of all Borel subalgebras of  $\mathfrak{g}$ ,  ${}^a\mathfrak{h}$  is the universal Cartan subalgebra. For any Borel subalgebra  $\mathfrak{b} \in \mathcal{B}$ , we have a canonical isomorphism:

$$\phi_{\mathfrak{b}} : {}^a\mathfrak{h} \simeq \mathfrak{h},$$

such that negative roots in  ${}^a\mathfrak{h}^*$  correspond to roots in  $\Delta(\mathfrak{h}, \mathfrak{n})$ .

We can define a  $G$ -equivariant vector bundle  $\mathcal{L}_{\lambda}$  over  $\mathcal{B}$  for any  $\lambda \in {}^a\mathfrak{h}^*$  by:

$$\mathcal{L}_{\lambda}|_{\mathfrak{b}} = \mathbb{C}_{\lambda} := \text{1-dimensional representation of } B \text{ defined by } \lambda \circ \phi_{\mathfrak{b}}^{-1}.$$

$$\boxed{\text{Weight lattice } \Lambda \subseteq {}^a\mathfrak{h}^*} \xleftarrow{1-1} \boxed{G\text{-equivariant line bundles on } \mathcal{B}}$$

Under this correspondence,  $\mathcal{L}_{\lambda}$  is ample if and only if  $\lambda$  is regular dominant.  $\mathcal{L}_{-2\rho}$  is the canonical line bundle  $\wedge^{\text{top}} T^* \mathcal{B}$  on  $\mathcal{B}$ .



21. 2025.11.19

**21.1. compute annihilator variety.**

$$\boxed{\text{Prim}_\nu(\mathbf{U}(\mathfrak{g}))} \xleftarrow{1-1} \boxed{\text{left cells in Coh}_\nu(\mathfrak{g}, \mathfrak{b})} \longrightarrow \boxed{\text{double cells in Coh}_\nu(\mathfrak{g}, \mathfrak{b})}$$

$$I \in \text{Prim}(\mathbf{U}(\mathfrak{g})) \longmapsto \mathcal{C}_{\nu, I}^L \longmapsto \mathcal{C}_{\nu, I}^{LR}$$

$\text{span}(\mathcal{C}_{\nu, I}^{LR})$  is a representation of the integral Weyl group  $W(\nu)$ , all irreducible constituents of  $\text{span}(\mathcal{C}_{\nu, I}^{LR})$  lies in a single double cell, and contains a unique special representation of  $W(\nu)$

$$\sigma_{\nu, I}.$$

Then by results of Joseph, we have:

$$\text{AV}(I) = \mathcal{O}(j_{W(\nu)}^W(\sigma_{\nu, I})).$$

Which means that to compute the annihilator variety of  $L(w\nu)$ , we only need to compute the corresponding cell representation.

In the case of  $(\mathfrak{g}, K)$ -modules, the method is the same:

$$\begin{array}{c} X: \text{Irreducible } (\mathfrak{g}, K)\text{-module} \\ \Downarrow \\ C: \text{Harish-Chandra cell containing } X \\ \Downarrow \\ \sigma_C: \text{special representation of } W(\nu) \text{ contained in the cell representation} \\ \Downarrow \\ \mathcal{O}(j_{W(\nu)}^W(\sigma_C)) = \text{AV}(X) \end{array}$$

**Remark 21.1.** The key point is the result of Joseph [Jos85] the Springer correspondence of cell representation gives the annihilator variety. However, I'm not fully understand the details.

**21.2. annihilator varieties are special.**  $j$ -induction of parabolic subgroups preserves special representations.

To prove that the annihilator variety of irreducible  $(\mathfrak{g}, K)$ -module is closure of a single special nilpotent orbit, only need to check the case when infinitesimal character has good parity and bad parity.

For example, in the case of  $G = \text{Sp}_{2n}(\mathbb{R})$ , we have:

$$\text{Coh}_{\Lambda_b}(\text{SO}(n_b, n_b)) \otimes \text{Coh}_{\Lambda_g}(\text{Sp}_{2n_g}(\mathbb{R})) \simeq \text{Coh}_{\Lambda}(\text{Sp}_{2n}(\mathbb{R})).$$

We need to show that the  $j$ -induction of any special repn appears in  $\text{Coh}_{\Lambda_b}(\text{SO}(n_b, n_b))$  and any special repn appears in  $\text{Coh}_{\Lambda_g}(\text{Sp}_{2n_g}(\mathbb{R}))$  gives a special repn in  $\text{Coh}_{\Lambda}(\text{Sp}_{2n}(\mathbb{R}))$ .

- **Problem:** How to compute the  $j$ -induction explicitly?
- **Problem:** the special repn of each cell always appears?

22. 2025.11.21

**Some problems/plans**

- Use counting methods to prove that all annihilator variety of irreducible  $(\mathfrak{g}, K)$ -modules of orthogonal or special groups are special.
- Use counting methods to prove that all annihilator variety of irreducible genuine  $(\mathfrak{g}, K)$ -modules of metaplectic groups are metaplectic special.
- By computation, prove that in the case of metaplectic groups, metaplectic quasi-admissible equivalent to metaplectic special. **solved! Yes**
- Prove that for metaplectic groups, whittaker support of irreducible genuine  $(\mathfrak{g}, K)$ -modules are metaplectic quasi-admissible. **Use arguments of Gomez-Gourevitch-Sah**
- Prove associated cycles coincides with whittaker cycles for irreducible  $(\mathfrak{g}, K)$ -modules of real classical groups and metaplectic groups.

## 23. 2025.11.24

For a Weyl group  $W$ , elements of  $W$  are participated into double cells, left cells, etc. Each double cell  $\mathcal{C}^{LR}$  contains a unique special representation  $\sigma_{\mathcal{C}^{LR}}$  of  $W$ . And each left cell  $\mathcal{C}^L$  corresponds to a primitive ideal  $I_{\mathcal{C}^L}$  of  $U(\mathfrak{g})$  with infinitesimal character  $\rho$ . For each element  $w \in W$ , there are two processes to get a nilpotent orbit  $\mathcal{O}_w$  in  $\mathfrak{g}^*$ :

$$w \longmapsto \mathcal{C}_w^{LR} \longmapsto \sigma_{\mathcal{C}_w^{LR}} \longmapsto \mathcal{O}(\sigma_{\mathcal{C}_w^{LR}});$$

$$w \xrightarrow{RS} P(w) \longmapsto \mathcal{O}_{P(w)} \longmapsto \mathcal{O}_{P(w),s}.$$

The question is: are these two nilpotent orbits the same? i.e.  $\mathcal{O}(\sigma_{\mathcal{C}_w^{LR}}) = \mathcal{O}_{P(w),s}$ ?

answer: Yes! [BV82]

## 24. 2026.3.3

$G$  is a linear real reductive group,  $\tilde{G}$  is a double cover of  $G$ .

**Theorem 24.1.**  *$\tilde{G}_0$  is a linear group (i.e. there is an injective homomorphism  $\tilde{G}_0 \rightarrow \mathrm{GL}_n(\mathbb{R})$  for some  $n$ ) equivalent to the existence of a finite-dimensional genuine representation of  $\tilde{G}$ .*

*Proof.* If  $\tilde{G}_0$  is a linear group, there are two possibilities:

- $\varepsilon \notin \tilde{G}_0$ : then  $\{1, \varepsilon\} \times \tilde{G}_0$  is a subgroup of  $\tilde{G}$  with finite index, and it has a finite-dimensional representation such that  $\varepsilon$  acts by  $-1$ , so we get a finite-dimensional genuine representation of  $\tilde{G}$  by induction.
- $\varepsilon \in \tilde{G}_0$ : denote  $\tilde{G}_0$  by  $H$ , then  $H = A \times {}^0H$  as a Nash group, and  ${}^0H$  is also a linear group. So there is a closed embedding of Nash groups  ${}^0H \hookrightarrow \mathrm{GL}_n(\mathbb{R})$  for some  $n$  ( ${}^0H$  has compact center), take Zariski closure of  ${}^0H$  in  $\mathrm{GL}_n(\mathbb{C})$  we get the complexification  ${}^0H_{\mathbb{C}}$  of  ${}^0H$ , there is an algebraic matrix coefficient of  ${}^0H_{\mathbb{C}}$  such that  $\varepsilon$  acts on it by  $-1$ , and it generates a finite-dimensional genuine representation of  $\tilde{G}$  by induction.

Conversely, if there is a finite-dimensional genuine representation of  $\tilde{G}$ , we also have two possibilities:

- $\varepsilon \notin \tilde{G}_0$ : then  $\tilde{G}_0 \rightarrow G$  is injective, since  $G$  is a linear group,  $\tilde{G}_0$  is also a linear group.
- $\varepsilon \in \tilde{G}_0$ : then we can find a finite-dimensional representation  $E_1$  of  $\tilde{G}_0$  such that  $\varepsilon$  acts by  $-1$ . We also have a finite-dimensional faithful representation  $E_2$  of  $G$ , we can regard it as a representation of  $\tilde{G}_0$  by pullback, and  $\varepsilon$  acts trivially on  $E_2$ . So  $E_1 \oplus E_2$  is a finite-dimensional faithful representation of  $\tilde{G}_0$ .

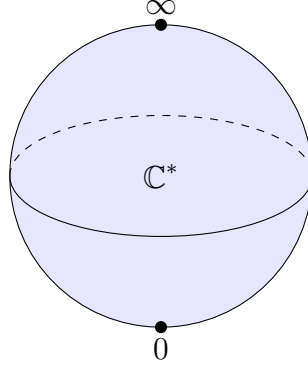
□

**Remark 24.2.** After I have a sleep, I find that  $\tilde{G}_0$  can be replaced by  $\tilde{G}$  in this theorem, and the proof is even easier.

## 25. 2026.3.21

In this section, we illustrate the geometry of the flag variety of  $\mathrm{SL}_2(\mathbb{C})$ , which serve as a simple example for understanding more general phenomena.

$K = \mathbb{C}^*$  action has three orbits on  $\mathbb{P}^1(\mathbb{C})$ : two closed orbits  $\{0\}$  and  $\{\infty\}$ , and one open orbit  $\mathbb{C}^*$ . The conormal bundle of the open orbit is the zero section, while the conormal bundle of the closed orbits are the cotangent fiber at 0 and  $\infty$  respectively. So we have three conormal bundles, which are all Lagrangian subvarieties in  $T^*\mathbb{P}^1(\mathbb{C})$ .

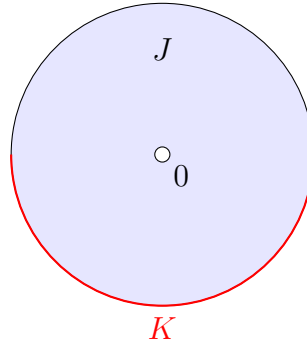


For a  $K$ -equivariant perverse sheaf  $\mathcal{F}^\bullet$  on it, we calculate its characteristic cycle  $\mathrm{CC}(\mathcal{F}^\bullet)$  use the formula

$$\chi_S^{\mathrm{mic}} = \sum_{S' > S} c(S, S') \chi_{S'}^{\mathrm{loc}},$$

only need to determine the coefficients  $c(S, S') (= \sum_i (-1)^i H_c^i(J \cap S', K \cap S'; \mathbb{C}))$  for  $S' > S$ , where  $(J, K)$  is the normal Morse datum of  $S$ , which is well-defined up to homeomorphism.

- For the open orbit  $\mathbb{C}^*$ , we have  $S' = \mathbb{C}^*$  and  $c(\mathbb{C}^*, \mathbb{C}^*) = -1$ .
- For the closed orbit  $\{0\}$ , we have  $S' = \{0\}, \mathbb{C}^*$ , and  $c(\{0\}, \{0\}) = 1$ , for  $c(\{0\}, \mathbb{C}^*)$ , the pair  $(J \cap S', K \cap S')$  is the following picture



therefore  $c(\{0\}, \mathbb{C}^*) = \sum_i (-1)^i H_c^i(J \cap \mathbb{C}^*, K \cap \mathbb{C}^*; \mathbb{C}) = ? -1$ .

So we have

$$\begin{aligned} \chi_0^{\mathrm{mic}} &= \chi_0^{\mathrm{loc}} - \chi_{\mathbb{C}^*}^{\mathrm{loc}}, \\ \chi_\infty^{\mathrm{mic}} &= \chi_\infty^{\mathrm{loc}} - \chi_{\mathbb{C}^*}^{\mathrm{loc}}, \\ \chi_{\mathbb{C}^*}^{\mathrm{mic}} &= -\chi_{\mathbb{C}^*}^{\mathrm{loc}}. \end{aligned}$$

## 26. 2026.3.22

Constructible sheaves on  $\mathbb{C} = \mathbb{C}^* \sqcup \{0\}$ .

Denote  $j : \mathbb{C}^* \hookrightarrow \mathbb{C}$  the open inclusion,  $i : \{0\} \hookrightarrow \mathbb{C}$  the closed inclusion. Then we have the following six functors:

$$\begin{array}{ccccc} & \xrightarrow{j_!} & & \xrightarrow{i^*} & \\ D^b(\mathbb{C}^*) & \xleftarrow{j^*=j^!} & D_c^b(\mathbb{C}) & \xleftarrow{i_*=i_!} & D^b(\{0\}) \\ & \xrightarrow{j_*} & & \xrightarrow{i^!} & \end{array}$$

$\mathcal{L}$  is a local system on  $\mathbb{C}^*$ .

$j_!\mathcal{L}$  has no quotient supported on  $\{0\}$ ,  $j_*$  has no subobject supported on  $\{0\}$ .  $j_!\mathcal{L}[1]$  and  $j_*\mathcal{L}[1]$  are perverse sheaves on  $\mathbb{C}$ .

**Remark 26.1.** Actually when  $j : S \hookrightarrow X$  is **affine locally closed embedding**,  $j_!$  and  $j_*$  are  $t$ -exact for the perverse  $t$ -structure, so  $j_!\mathcal{L}[\dim S]$  and  $j_*\mathcal{L}[\dim S]$  are perverse sheaves on  $X$ .

By easy calculation of group cohomology, we have:

$$H^i(j_*\mathcal{L}[1]_0) = \begin{cases} 0 & i \neq 0, -1 \\ V_\mu & i = 0 \\ V^\mu & i = -1 \end{cases}$$

There is an exact sequence of perverse sheaves on  $\mathbb{C}$ :

$$0 \rightarrow i_*V^\mu \rightarrow j_!\mathcal{L}[1] \rightarrow j_*\mathcal{L}[1] \rightarrow i_*V_\mu \rightarrow 0.$$

27. 2026.3.31

## I. The Enhanced Flag Variety and $G_0/N$ Isomorphism [BB93]

$\mathfrak{g}$  be a complex semisimple Lie algebra,  ${}^a\mathfrak{h} = \mathfrak{b}/[\mathfrak{b}, \mathfrak{b}]$  be the universal Cartan subalgebra,  $G$  be the automorphism group of  $\mathfrak{g}$ ,  $G_0$  be the connected component of  $G$ , it is the adjoint group of  $\mathfrak{g}$ .

Attached to  ${}^a\mathfrak{h}$  is a root system  $\Delta$ , and the Weyl group  $W$ . Although there isn't a well defined action of  ${}^a\mathfrak{h} = \mathfrak{b}/[\mathfrak{b}, \mathfrak{b}]$  on  $\mathfrak{g}/\mathfrak{b} = \mathfrak{n}^*$ , the action of  ${}^a\mathfrak{h}$  on  $(\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])^*$  is well defined, and there is a canonical decomposition:

$$(\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])^* = \bigoplus_{\alpha \in \Sigma} (\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])_{\alpha}^*,$$

Where  $\Sigma$  is the set of simple roots. Note that this choice of positive root system is opposite to the one usually used in representation theory, but it is more natural in geometry since dominant implies positive line bundle in this way.

The enhanced flag variety  $\tilde{\mathcal{B}}$  is defined as the variety of pairs  $(\mathfrak{b}, \{x_{\alpha}\}_{\alpha \in \Sigma})$  where  $\mathfrak{b}$  is a Borel subalgebra of  $\mathfrak{g}$ , and  $\xi$  is a nonzero element in  $(\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])_{\alpha}^*$ . There is a natural projection map  $\pi : \tilde{\mathcal{B}} \rightarrow \mathcal{B}$  which forgets the second component, and is an  $H$  principle bundle.

There is a  $G$  action on  $\tilde{\mathcal{B}}$  defined by  $g \cdot (\mathfrak{b}, \{x_{\alpha}\}_{\alpha \in \Sigma}) = (\text{Ad}(g)\mathfrak{b}, \{\text{Ad}^*(g)x_{\alpha}\}_{\alpha \in \Sigma})$ . There is also an  $H$  action on  $\tilde{\mathcal{B}}$  defined by  $h \cdot (\mathfrak{b}, \{x_{\alpha}\}_{\alpha \in \Sigma}) = (\mathfrak{b}, \{\alpha(h)x_{\alpha}\}_{\alpha \in \Sigma})$ . These two actions are compatible in the following way:

$$g \cdot h \cdot (\mathfrak{b}, \{x_{\alpha}\}_{\alpha \in \Sigma}) = \kappa_g(h) \cdot g \cdot (\mathfrak{b}, \{x_{\alpha}\}_{\alpha \in \Sigma}),$$

where  $\kappa : G \rightarrow \text{Aut}(H)$  is the action as outer automorphisms of  $H$ .

The adjoint group  $G_0$  acts transitively on  $\tilde{\mathcal{B}}$ , and the stabilizer of a base point  $(\mathfrak{b}, \{x_{\alpha}\}_{\alpha \in \Sigma})$  is  $N$ . So we have an isomorphism  $\tilde{\mathcal{B}} \simeq G_0/N$ . Under this identification, the  $G$  action on  $\tilde{\mathcal{B}}$  is given by  $g \cdot (g_0N) = gg_0N$ , and the  $H$  action on  $\tilde{\mathcal{B}}$  is given by  $h \cdot (g_0N) = g_0hN$ .

## II. Line Bundles on $\mathcal{B}$ and Very Ampleness

Now  $G$  is a connected reductive group.

**Lemma 27.1.** *Let  $G \rightarrow \text{GL}(V)$  be a finite-dimensional representation of  $G$ ,  $k \cdot v \subseteq V$  be a one-dimensional subspace,  $P = \text{Stab}_G(k \cdot v)$ . Suppose that  $k_{\lambda}$  be the one-dimensional representation of  $P$  defined by the action of  $P$  on  $k \cdot v$ , then  $i^*(\mathcal{O}_{\mathbb{P}(V)}(1)) = \mathcal{O}_{G/P}(\lambda)$ .*

*Proof.* Note that  $\mathcal{O}_{\mathbb{P}(V)}(-1)$  is the tautological line bundle,  $P$  action on the fiber of  $\mathcal{O}_{\mathbb{P}(V)}(-1)$  at  $k \cdot v$  is given by  $k_{\lambda}$ . So  $k_{-\lambda}$  for  $\mathcal{O}(1)$ , also note that  $\mathcal{O}_{G/P}(\lambda) = G \times^P k_{-\lambda}$ .  $\square$

**Lemma 27.2.**  *$\mathcal{O}_{\mathcal{B}}(\lambda)$  is very ample if and only if  $\lambda$  is regular dominant ( $\lambda \in X^+(H)$ ).*

*Proof.* Consider the Verma module  $V^{\lambda} = \mathcal{U}(\mathfrak{g}) \otimes_{\mathcal{U}(\mathfrak{h})} k_{\lambda}$ . By the transitive group action of  $G$  on  $\mathcal{B}$ ,  $\mathcal{O}(\lambda)$  is generated by global sections  $\Gamma(\mathcal{B}, \mathcal{O}(\lambda)) = V^{\lambda}$ . Hence there is a surjection  $\Gamma(\mathcal{B}, \mathcal{O}(\lambda)) \twoheadrightarrow \mathcal{O}(\lambda) \otimes_k k_{eB} = k_{\lambda}$ . Taking dual, we have  $k_{-\lambda} \hookrightarrow (V^{\lambda})^*$ .

Since  $\lambda$  is regular dominant, the stabilizer of  $k_{-\lambda}$  in  $(V^{\lambda})^*$  is the Borel  $B$ , by the previous lemma, we have a  $G$ -equivariant morphism  $\mathcal{B} \hookrightarrow \mathbb{P}((V^{\lambda})^*)$  such that  $\mathcal{O}(\lambda)$  is the pullback of  $\mathcal{O}(1)$ , hence  $\mathcal{O}(\lambda)$  is very ample.  $\square$

## 28. 2026.4.5

$G$  is a complex algebraic group,  $P$  is a closed subgroup of  $G$  with a finite dimensional rational representation  $V$ . suppose that  $\mathfrak{g} = \mathfrak{u} \oplus \mathfrak{p}$ .

$$\begin{array}{ccc} Y = G \times^P V & \xrightarrow{\pi} & V \\ \downarrow & & \\ G/P & & \end{array}$$

There is an isomorphism:

$$\mathfrak{u} \oplus V \xrightarrow{\sim} T_{\overline{(1, \alpha)}}(G \times^P V),$$

which sends  $(u, v)$  to  $u_Y + v$ , where  $v \in V$  is regarded as a tangent vector of the fiber  $V$  at  $\alpha$ , and  $u_Y$  is the vector field on  $Y$  generated by the action of  $u \in \mathfrak{u} \subset \mathfrak{g}$ .

The tangent map of  $\pi$  at  $\overline{(1, \alpha)}$  is:

$$\begin{aligned} d\pi_{\overline{(1, \alpha)}} : \mathfrak{u} \oplus V &\rightarrow T_{\overline{(1, \alpha)}}(G \times^P V) \rightarrow V, \\ d\pi_{\overline{(1, \alpha)}}(u, v) &= u \cdot \alpha + v, \end{aligned}$$

since  $\pi(g, \alpha) = g \cdot \alpha$ .

The above calculation is instrumental in extending the pullback of the Kostant-Kirillov symplectic form to the Jacobson-Morozov resolution of a nilpotent orbit:

$$\begin{array}{ccc} G \times^P \mathfrak{g}_{\geq 2} & \xrightarrow{\pi} & \overline{\mathbb{O}} \\ \uparrow & & \uparrow \\ G \times^P P \cdot e = G/Z_P(e) & \longrightarrow & \mathbb{O} \end{array}$$

$$\pi^* \omega_{KK, \alpha}((u, v), (x, y)) = \kappa(\alpha, [u, x]) + \kappa(u, y) - \kappa(v, x).$$



## REFERENCES

- [ABV92] Jeffrey Adams, Dan Barbasch, and David A. Vogan Jr. *The Langlands classification and irreducible characters for real reductive groups*. Vol. 104. Progress in Mathematics. Birkhäuser Boston, Inc., Boston, MA, 1992, pp. xii+318. MR1162533. DOI: [10.1007/978-1-4612-0383-4](https://doi.org/10.1007/978-1-4612-0383-4) (↑ 34).
- [BB82] Walter Borho and Jean-Luc Brylinski. “Differential operators on homogeneous spaces. I. Irreducibility of the associated variety for annihilators of induced modules”. In: *Invent. Math.* 69.3 (1982), pp. 437–476. MR679767. DOI: [10.1007/BF01389364](https://doi.org/10.1007/BF01389364) (↑ 13).
- [BB93] A. Beilinson and J. Bernstein. “A proof of Jantzen conjectures”. In: *I. M. Gel'fand Seminar*. Vol. 16, Part 1. Adv. Soviet Math. Amer. Math. Soc., Providence, RI, 1993, pp. 1–50. MR1237825 (↑ 45).
- [BV80] Dan Barbasch and David A. Vogan Jr. “The local structure of characters”. In: *J. Functional Analysis* 37.1 (1980), pp. 27–55. MR576644. DOI: [10.1016/0022-1236\(80\)90026-9](https://doi.org/10.1016/0022-1236(80)90026-9) (↑ 18).
- [BV82] Dan Barbasch and David Vogan. “Primitive ideals and orbital integrals in complex classical groups”. In: *Math. Ann.* 259.2 (1982), pp. 153–199. MR656661. DOI: [10.1007/BF01457308](https://doi.org/10.1007/BF01457308) (↑ 41).
- [GGS21] Raul Gomez, Dmitry Gourevitch, and Siddhartha Sahi. “Whittaker supports for representations of reductive groups”. In: *Ann. Inst. Fourier (Grenoble)* 71.1 (2021), pp. 239–286. MR4275869. DOI: [10.5802/aif.3372](https://doi.org/10.5802/aif.3372) (↑ 17).
- [GZ14] Raul Gomez and Chen-Bo Zhu. “Local theta lifting of generalized Whittaker models associated to nilpotent orbits”. In: *Geom. Funct. Anal.* 24.3 (2014), pp. 796–853. MR3213830. DOI: [10.1007/s00039-014-0276-5](https://doi.org/10.1007/s00039-014-0276-5) (↑ 18).
- [HTT08] Ryoshi Hotta, Kiyoshi Takeuchi, and Toshiyuki Tanisaki. *D-modules, perverse sheaves, and representation theory*. Japanese. Vol. 236. Progress in Mathematics. Birkhäuser Boston, Inc., Boston, MA, 2008, pp. xii+407. MR2357361. DOI: [10.1007/978-0-8176-4523-6](https://doi.org/10.1007/978-0-8176-4523-6) (↑ 29).
- [Jos85] Anthony Joseph. “On the associated variety of a primitive ideal”. In: *J. Algebra* 93.2 (1985), pp. 509–523. MR786766. DOI: [10.1016/0021-8693\(85\)90172-3](https://doi.org/10.1016/0021-8693(85)90172-3) (↑ 13, 39).
- [KL80] David Kazhdan and George Lusztig. “A topological approach to Springer’s representations”. In: *Adv. in Math.* 38.2 (1980), pp. 222–228. MR597198. DOI: [10.1016/0001-8708\(80\)90005-5](https://doi.org/10.1016/0001-8708(80)90005-5) (↑ 22).
- [KT84] M. Kashiwara and T. Tanisaki. “The characteristic cycles of holonomic systems on a flag manifold related to the Weyl group algebra”. In: *Invent. Math.* 77.1 (1984), pp. 185–198. MR751138. DOI: [10.1007/BF01389142](https://doi.org/10.1007/BF01389142) (↑ 21, 23).
- [KV95] Anthony W. Knap and David A. Vogan Jr. *Cohomological induction and unitary representations*. Vol. 45. Princeton Mathematical Series. Princeton University Press, Princeton, NJ, 1995, pp. xx+948. MR1330919. DOI: [10.1515/9781400883936](https://doi.org/10.1515/9781400883936) (↑ 13).
- [MW87] C. Mœglin and J.-L. Waldspurger. “Modèles de Whittaker dégénérés pour des groupes  $p$ -adiques”. In: *Math. Z.* 196.3 (1987), pp. 427–452. MR913667. DOI: [10.1007/BF01200363](https://doi.org/10.1007/BF01200363) (↑ 17).

- [Nev99] Monica Nevins. “Admissible nilpotent coadjoint orbits of  $p$ -adic reductive Lie groups”. In: *Represent. Theory* 3 (1999), pp. 105–126. MR1698202. DOI: [10.1090/S1088-4165-99-00072-2](https://doi.org/10.1090/S1088-4165-99-00072-2) (↑ 27).
- [Sch87] James Oliver Schwartz. *THE DETERMINATION OF THE ADMISSIBLE NILPOTENT ORBITS IN REAL CLASSICAL GROUPS*. Thesis (Ph.D.)—Massachusetts Institute of Technology. ProQuest LLC, Ann Arbor, MI, 1987, (no paging). MR2941135 (↑ 27).
- [SGA3] Michel Demazure and Alexander Grothendieck. *Schémas en Groupes (SGA 3). Séminaire de Géométrie Algébrique du Bois Marie 1962/64*. French. Lecture Notes in Mathematics 151, 152, 153. Tomes I, II, III. Berlin-New York: Springer-Verlag, 1970 (↑ 14).
- [Sun15] Binyong Sun. “Almost linear Nash groups”. In: *Chinese Ann. Math. Ser. B* 36.3 (2015), pp. 355–400. MR3341164. DOI: [10.1007/s11401-015-0915-7](https://doi.org/10.1007/s11401-015-0915-7) (↑ 17).